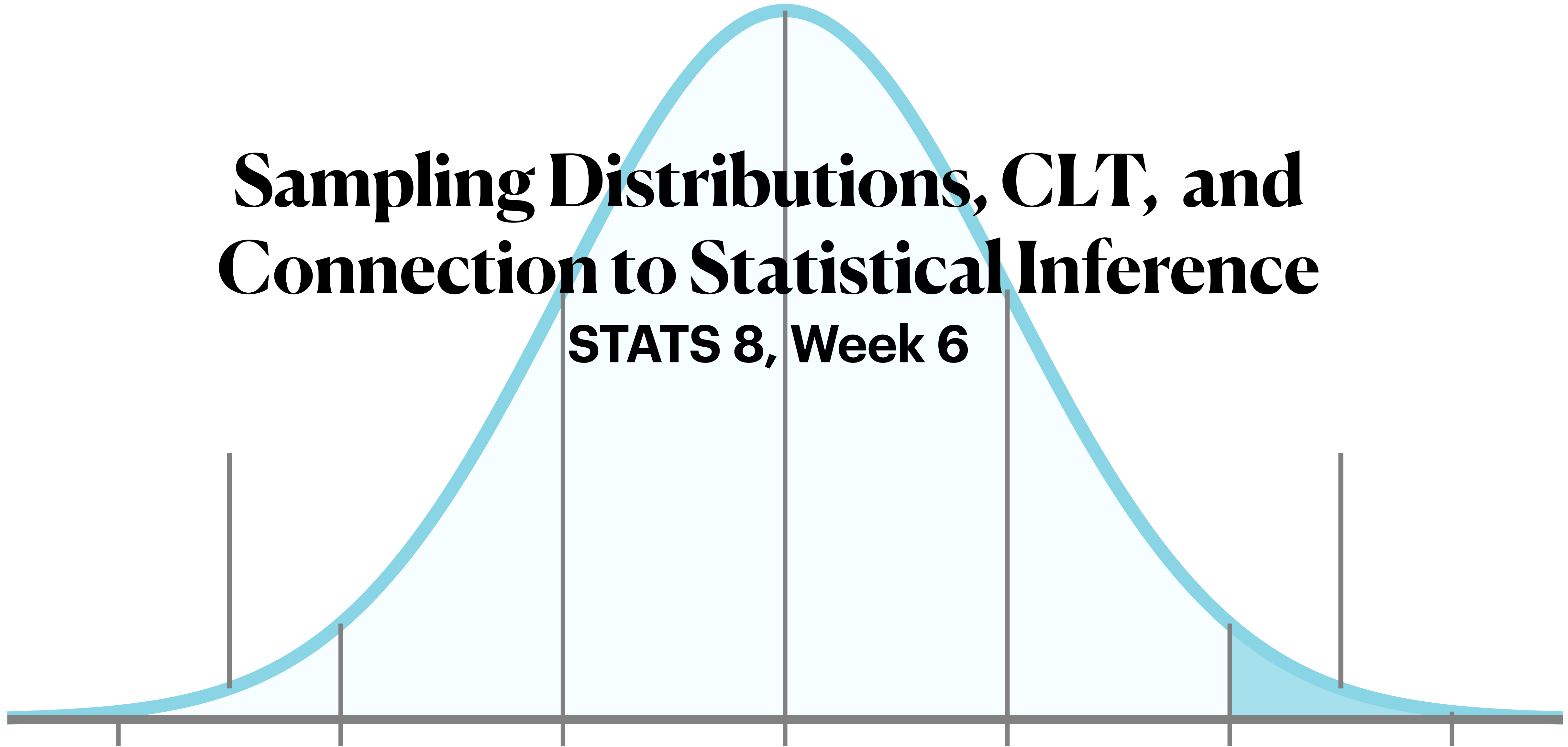


Sampling Distributions, CLT, and Connection to Statistical Inference

STATS 8, Week 6



Sampling from a Normal Distribution

$$X \sim N(98.6, 0.5^2)$$

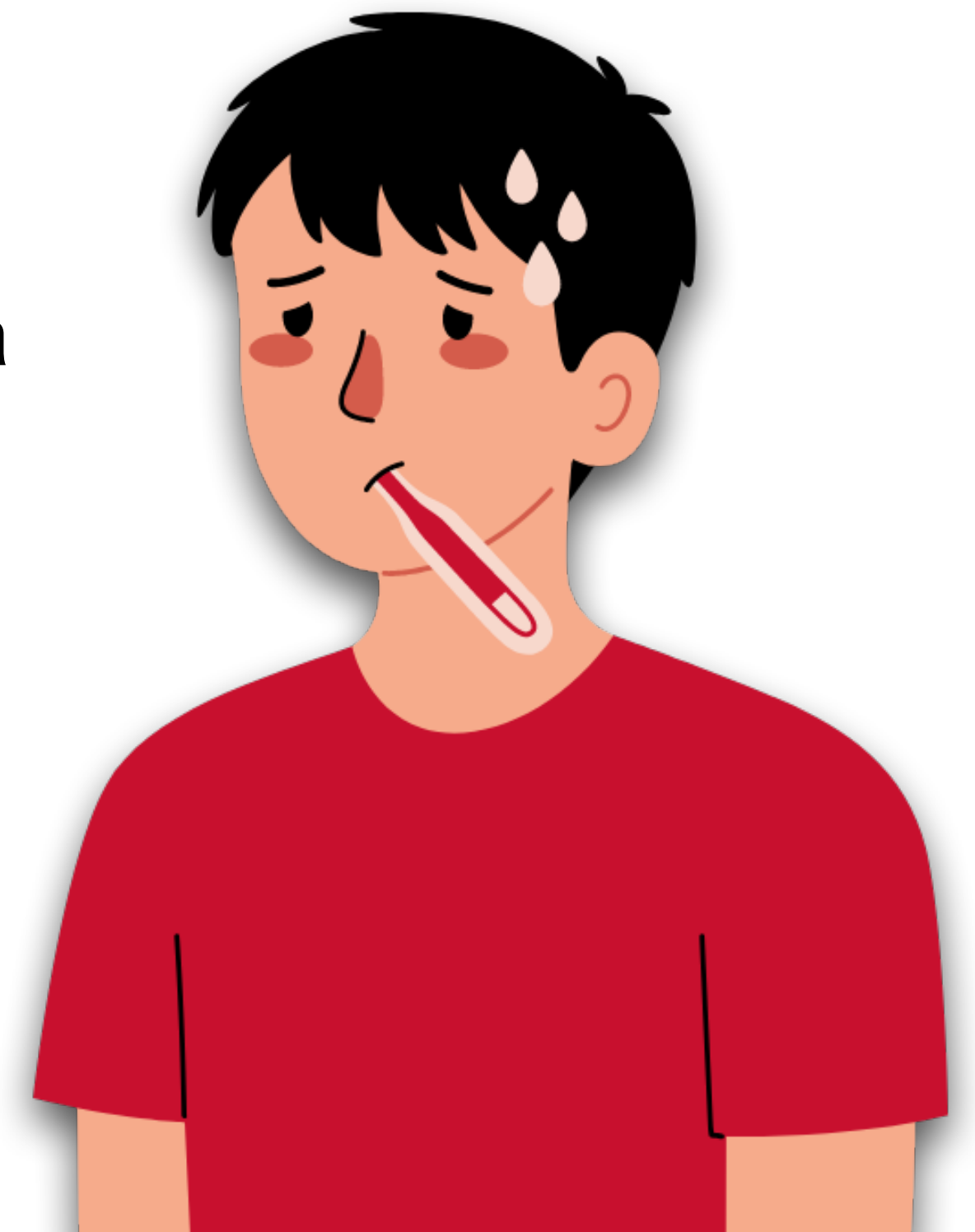
- Suppose that in the adult U.S. population, body temperature has a normal distribution with mean 98.6 degrees Fahrenheit and standard deviation 0.5 degrees.
- Note that $E(X) = 98.6$ is a population **parameter** because it is the average within the entire population of interest
 - Notation: from here forward we will use $\mu \equiv E(X)$, the population mean
- Let X = the body temperature of a randomly selected U.S. adult.
- What is $P(X \geq 100.4)$? (This is the temperature of a fever)
 - [PollEv.com/stats8](https://www.pollenv.com/stats8)



Sampling from a Normal Distribution

$$X \sim N(98.6, 0.5^2)$$

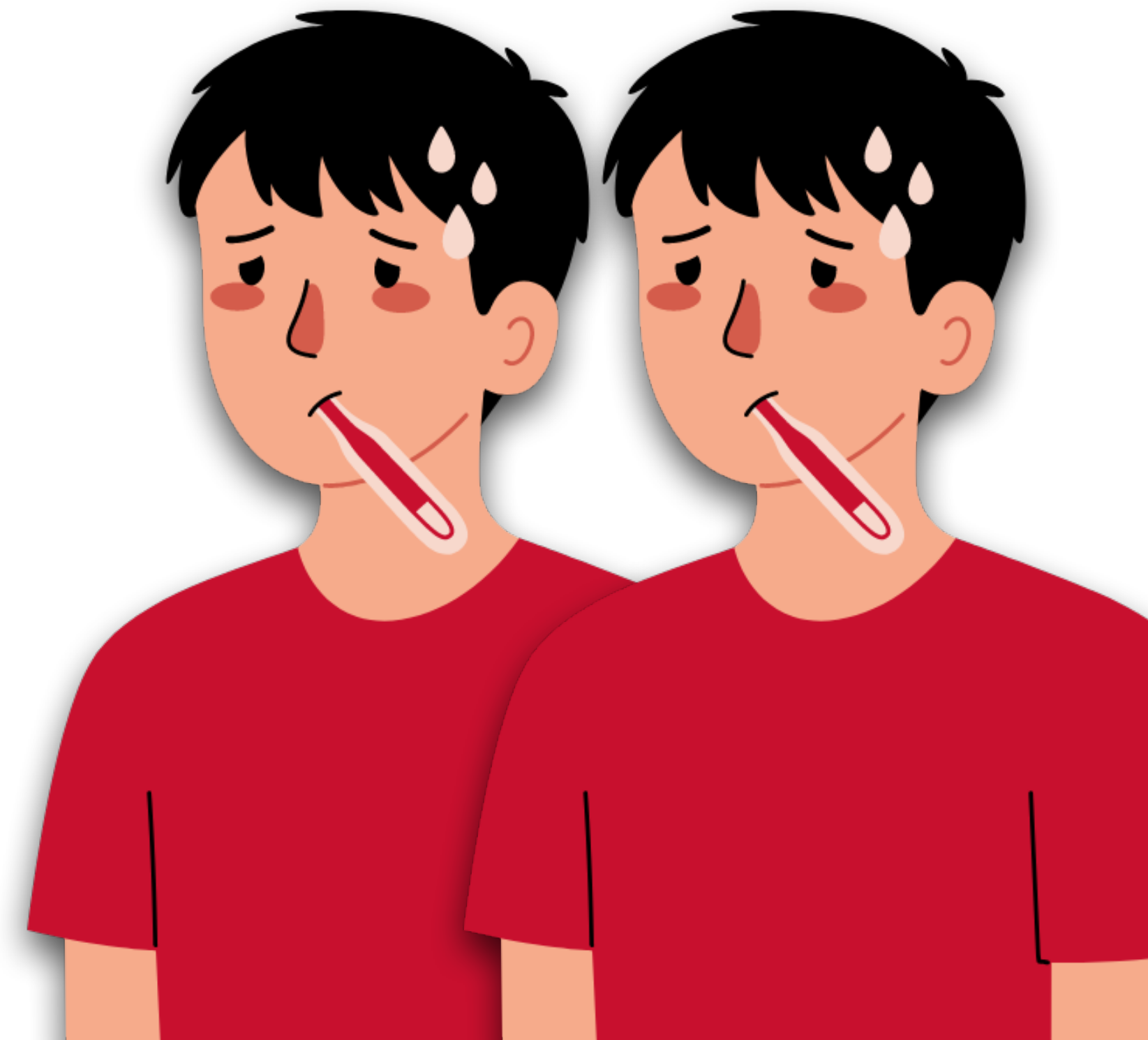
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- Let X = the body temperature of a randomly selected U.S. adult.
- What is $P(X \geq 100.4)$? (This is the temperature of a fever)
 - $1 - \text{pnorm}(100.4, \text{mean} = 98.6, \text{sd} = 0.5) = \mathbf{0.0002}$



Sampling from a Normal Distribution

$$X \sim N(98.6, 0.5^2)$$

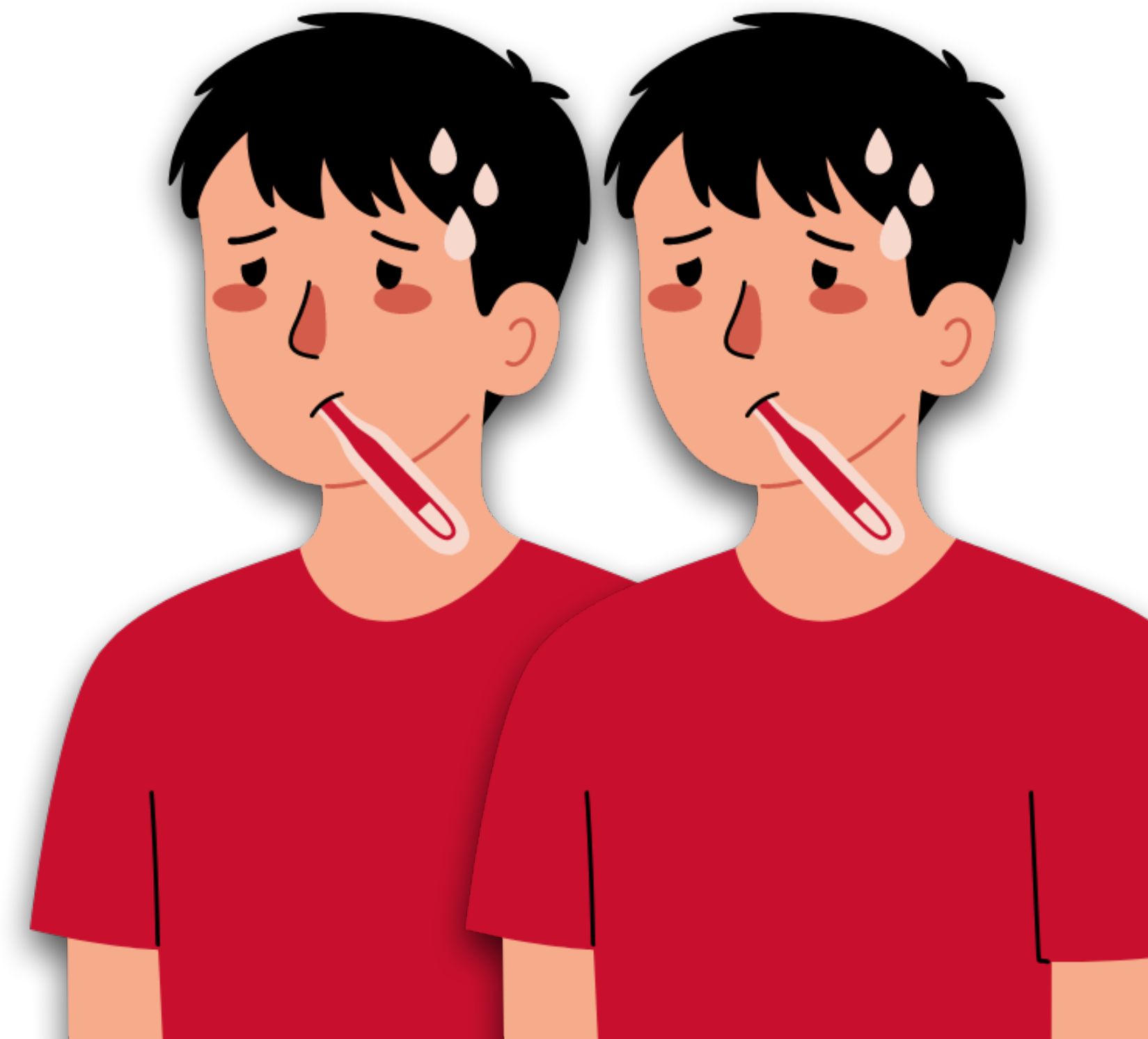
- Suppose that in the adult U.S. population, body temperature has a normal distribution with mean $\mu = 98.6$ degrees Fahrenheit and standard deviation 0.5 degrees.
- Now, instead of sampling only one person from the population, I randomly sample two people independently.
 - Let X_1 = the body temperature of the first randomly sampled person.
 - Let X_2 = the body temperature of the second randomly sampled person.
- $X_1 \sim N(98.6, 0.5^2)$, and also $X_2 \sim N(98.6, 0.5^2)$



Sampling from a Normal Distribution

$$X \sim N(98.6, 0.5^2)$$

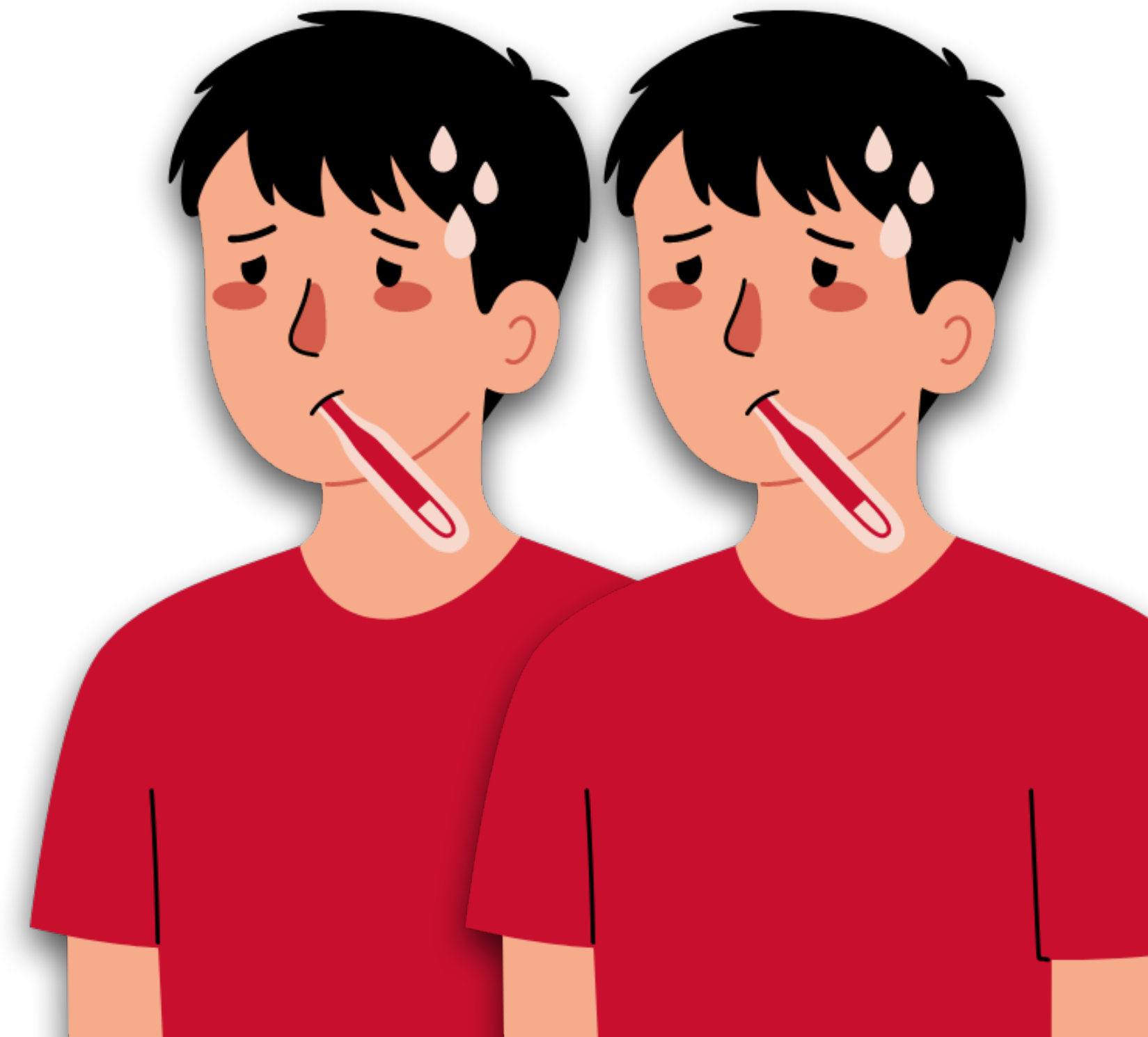
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- Now, instead of sampling only one person from the population, I randomly sample two people independently.
 - $X_1 \sim N(98.6, 0.5^2)$, and also $X_2 \sim N(98.6, 0.5^2)$
- Define a new quantity:
 - $\bar{X} = \frac{X_1 + X_2}{2}$ (the average of the two temperatures)
 - Is $\bar{X}$ a random variable? Why?
 - PollEv.com/stats8



Sampling from a Normal Distribution

$$X \sim N(98.6, 0.5^2)$$

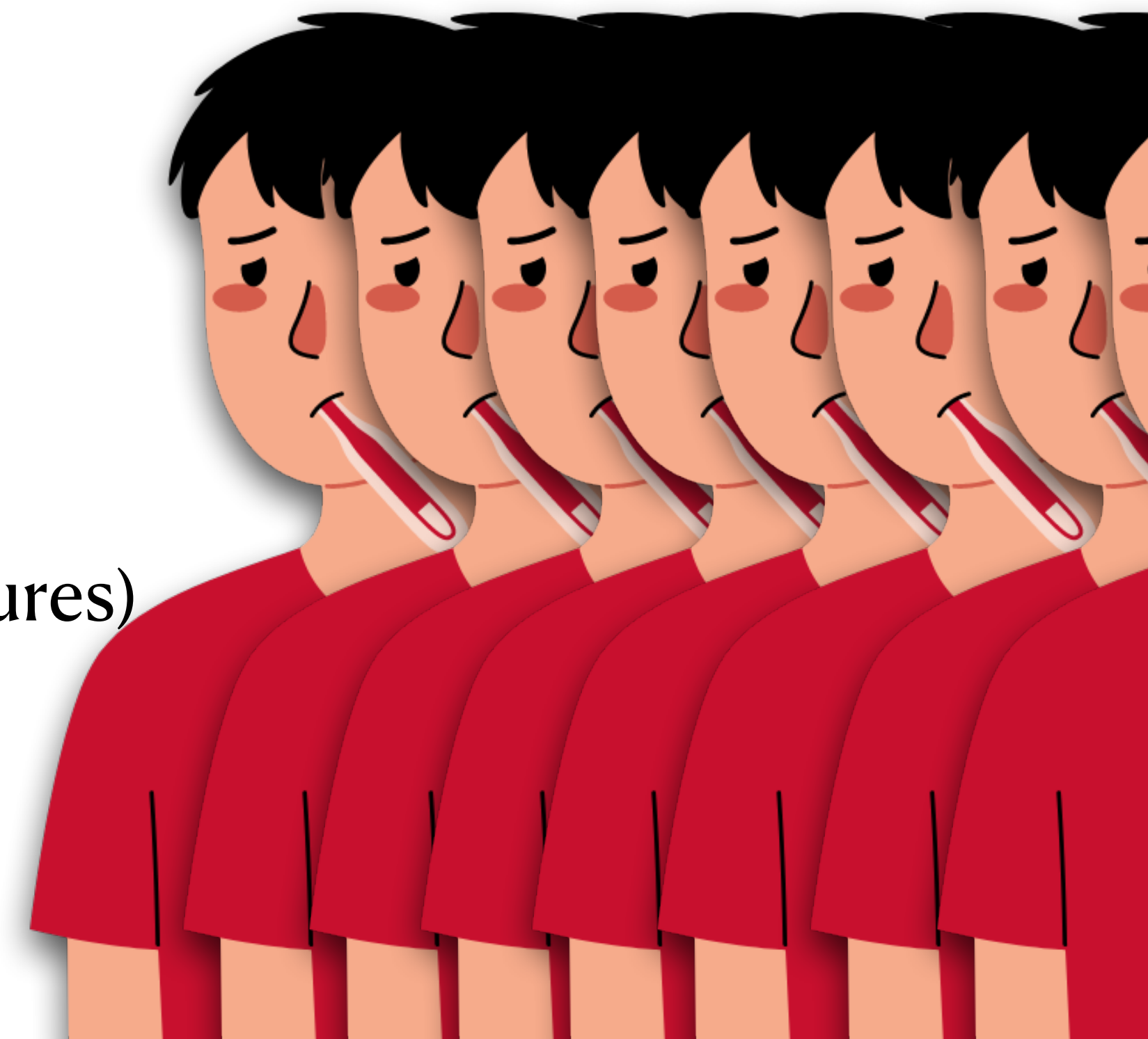
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 - $X_1 \sim N(98.6, 0.5^2)$, and also $X_2 \sim N(98.6, 0.5^2)$
- Define a new quantity:
 - $\bar{X} = \frac{X_1 + X_2}{2}$ (the average of the two temperatures)
 - Is $\bar{X}$ a random variable? Why?
 - Yes — it is computed from random quantities!



Sampling from a Normal Distribution

$$X \sim N(98.6, 0.5^2)$$

- Suppose that in the adult U.S. population, body temperature has a normal distribution with mean $\mu = 98.6$ degrees Fahrenheit and standard deviation 0.5 degrees.
- I can sample 3, or 4, or 5, so on...
 - $X_1 \sim N(98.6, 0.5^2), X_2 \sim N(98.6, 0.5^2), X_3 \sim N(98.6, 0.5^2),$
 $X_4 \sim N(98.6, 0.5^2), \dots, X_n \sim N(98.6, 0.5^2)$
- Define a new quantity:
 - $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ (the average of all the sampled temperatures)
 - $\bar{X}$ is still a random quantity!



PollEv.com/stats8

What is random?

- Suppose we are planning to randomly sample 5 U.S. adults and compute their average body temperature, $\bar{X}$. Is $\bar{X}$ a random variable?
 - A. Yes, because different random samples could give different sample means
 - B. Yes, because the population mean is unknown
 - C. No, because an average is not random, it's just a number
 - D. No, because no matter how you pick the sample, $\bar{X}$ will always be the same

PollEv.com/stats8

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PollEv.com/stats8

What is random?

- Now suppose we already selected the 5 people and computed their average body temperature as 98.46. Is this value random?
 - A. Yes, because it came from a random sample originally
 - B. Yes, because body temperature varies between people
 - C. No, because after the sample is selected and the value is computed, it is a fixed number
 - D. No, because sample means are not random quantities

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What is random?

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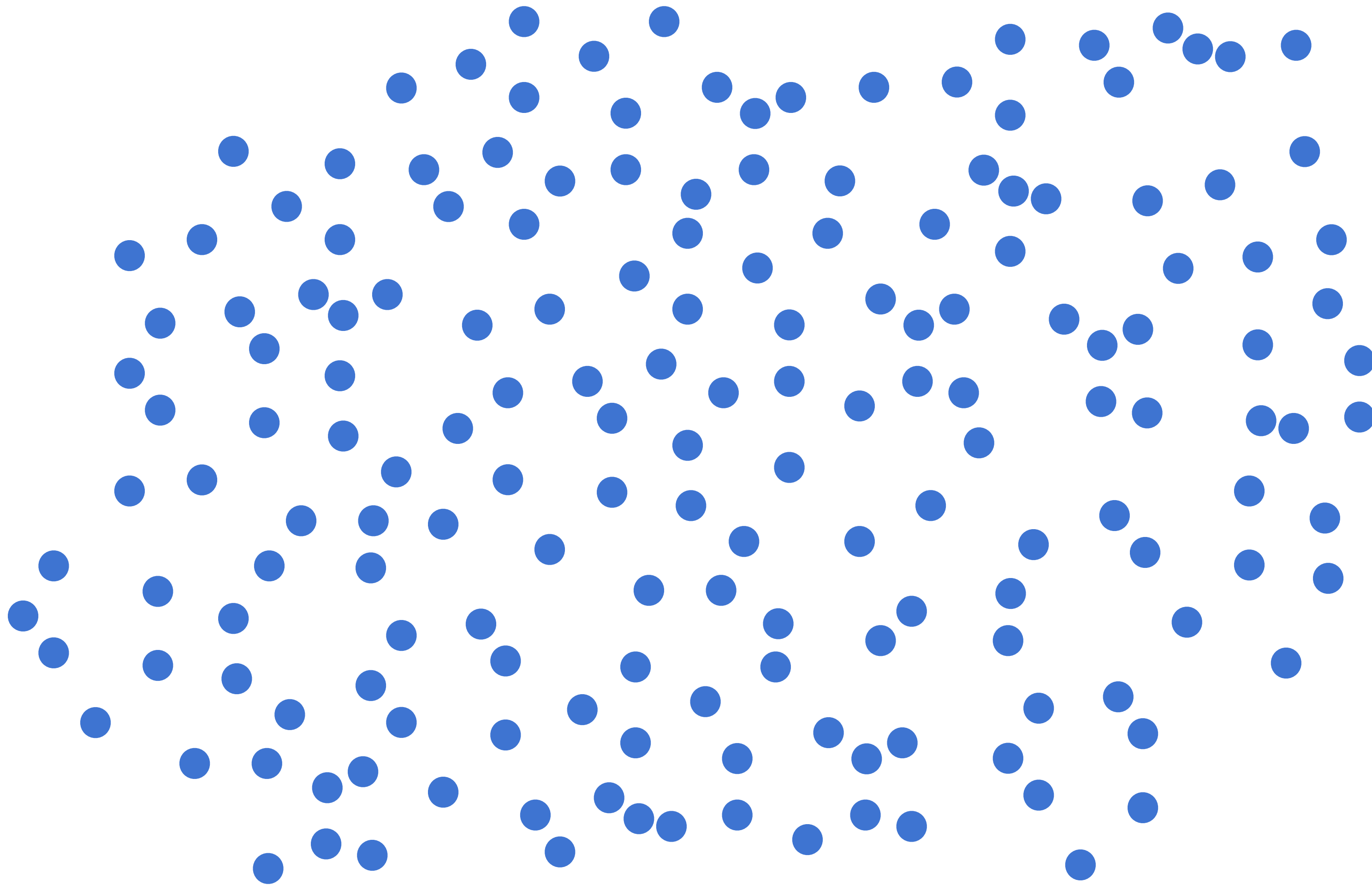
Sampling Distribution

Of a sample mean, $\bar{X}$

- We usually only get one sample. But to understand how trustworthy that one sample is, we need to imagine what could have happened across many possible random samples
- If $\bar{X}$ is a random variable, it must have a probability distribution
 - We call the distribution of $\bar{X}$ the **sampling distribution** of the **sample mean**
 - $\bar{X}$ is the **sample mean** (it is the average of the randomly sampled values)
 - A **sampling distribution** for a statistic is the probability distribution of possible values of the statistic for repeated samples of the same size taken from the same population
- What does this mean in practice?

Sampling Distribution

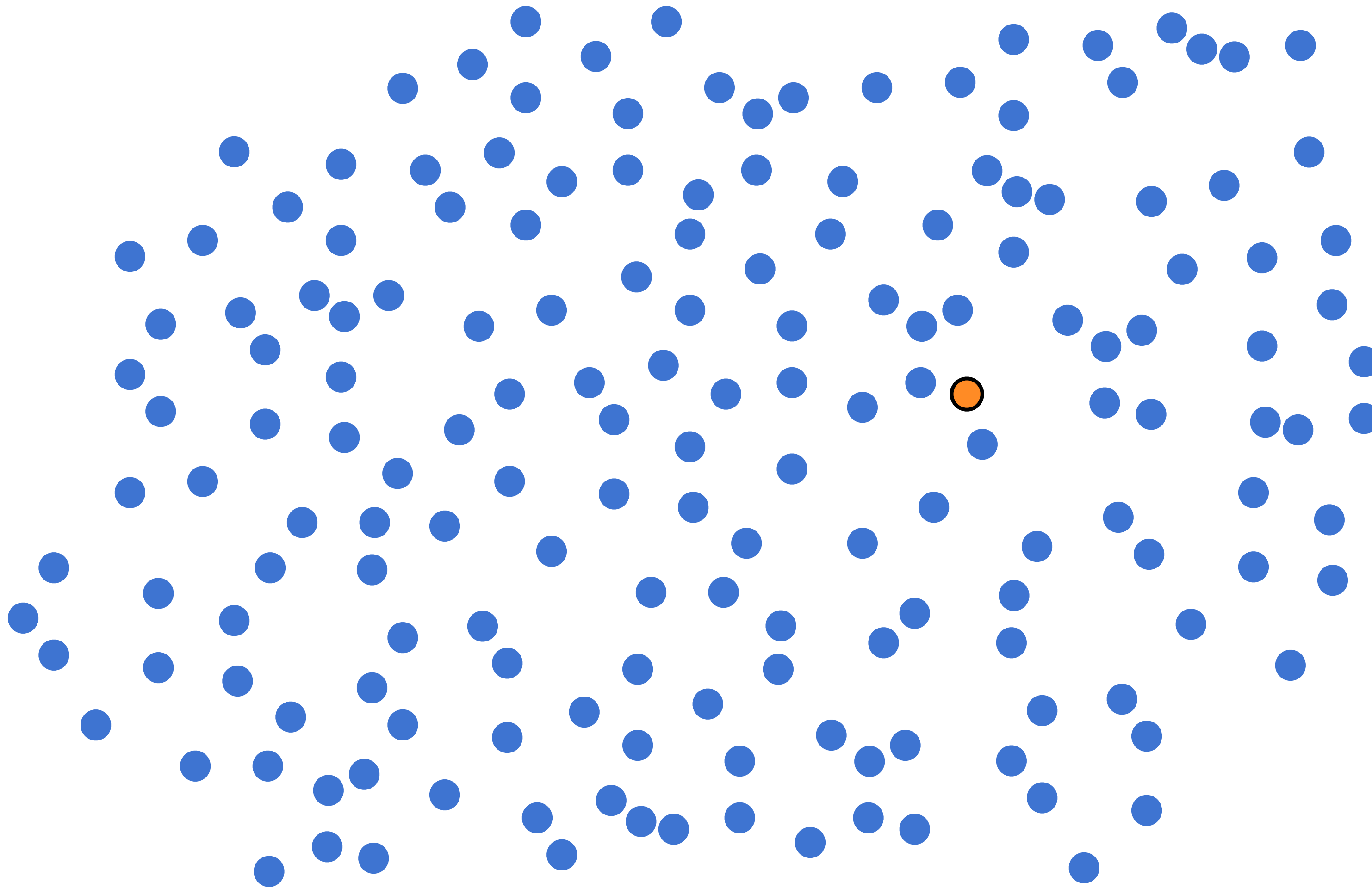
Of a sample mean, $\bar{X}$



- Imagine that the dots on the left represent a population of interest, where each dot represents a person
 - e.g., the population of U.S. adults or the population of UCI students
- The population mean body temperature of all the dots is $\mu = 98.6$

Sampling Distribution

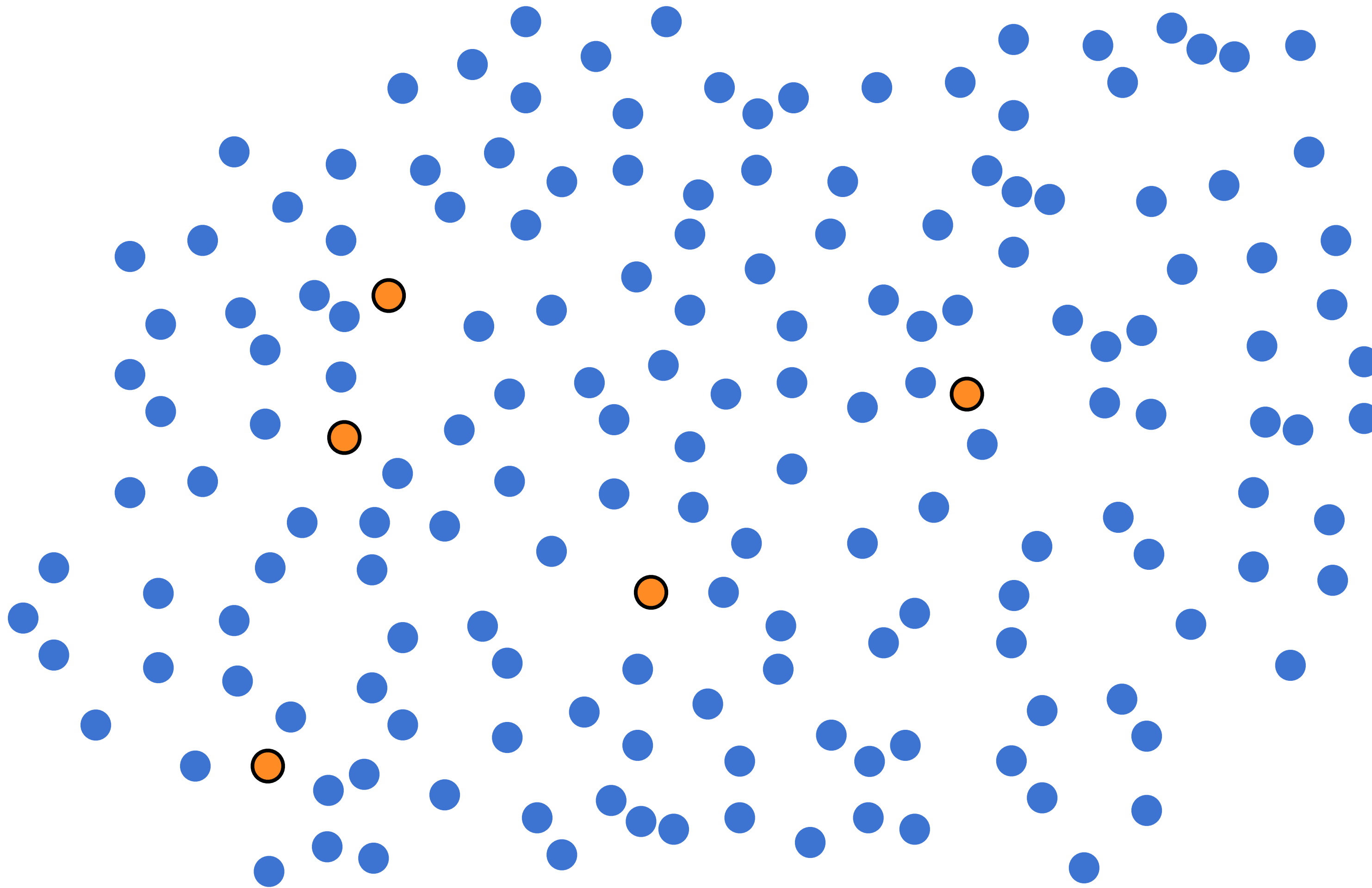
Of a sample mean, $\bar{X}$



- Imagine that the dots on the left represent a population of interest, where each dot represents a person
 - $\mu = 98.6$
- We can sample a person randomly and take their body temperature
 - e.g., 99.2

Sampling Distribution

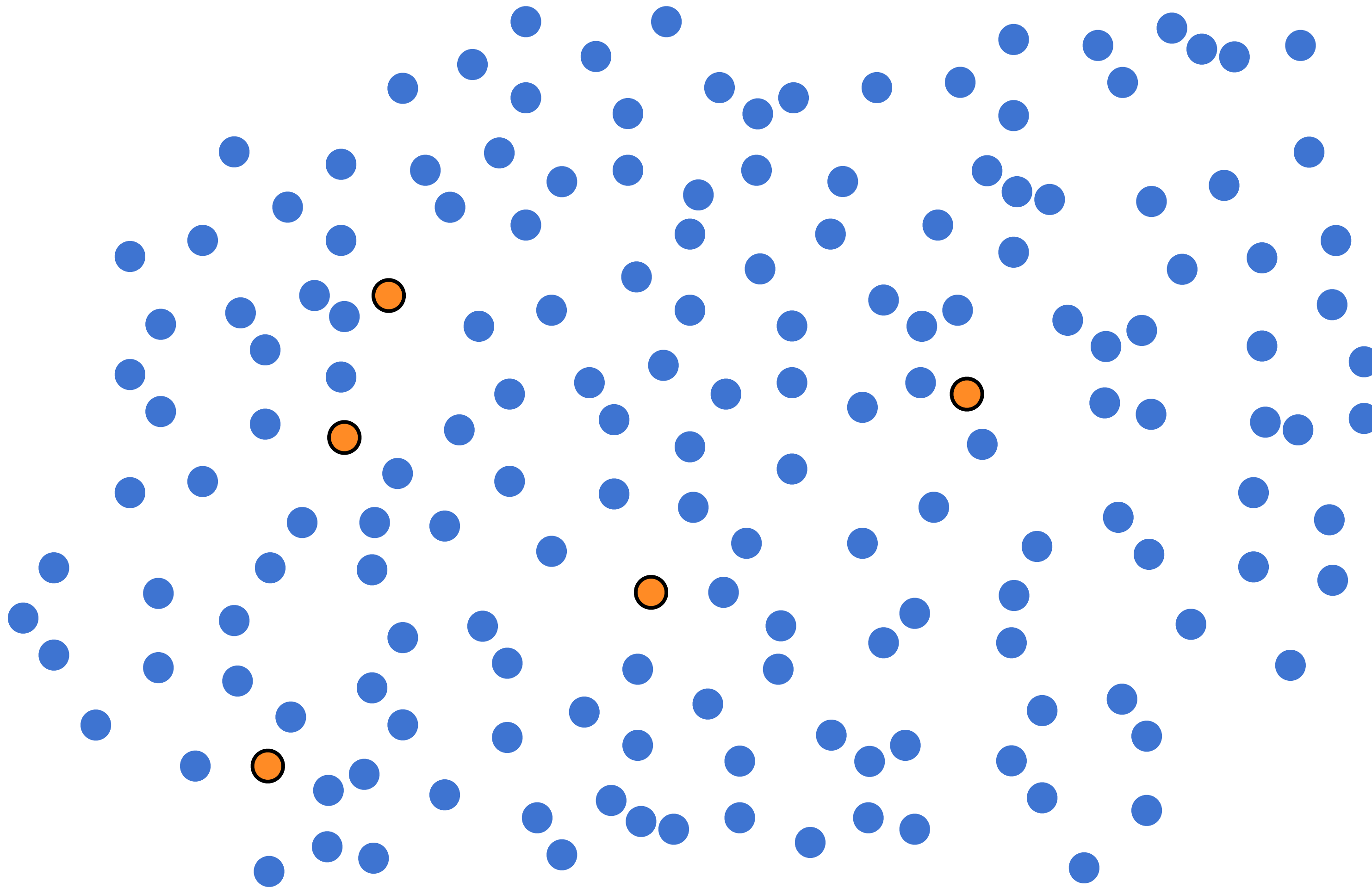
Of a sample mean, $\bar{X}$



- We can also randomly sample many people, and take their temperatures
- $\mu = 98.6$
- e.g., $X_1 = 99.2, X_2 = 98.6, X_3 = 99.1, X_4 = 97.4, X_5 = 98.0$

Sampling Distribution

Of a sample mean, $\bar{X}$



- We can also randomly sample many people, and take their temperatures

- $\mu = 98.6$

- e.g., 99.2, 98.6, 99.1, 97.4, 98.0

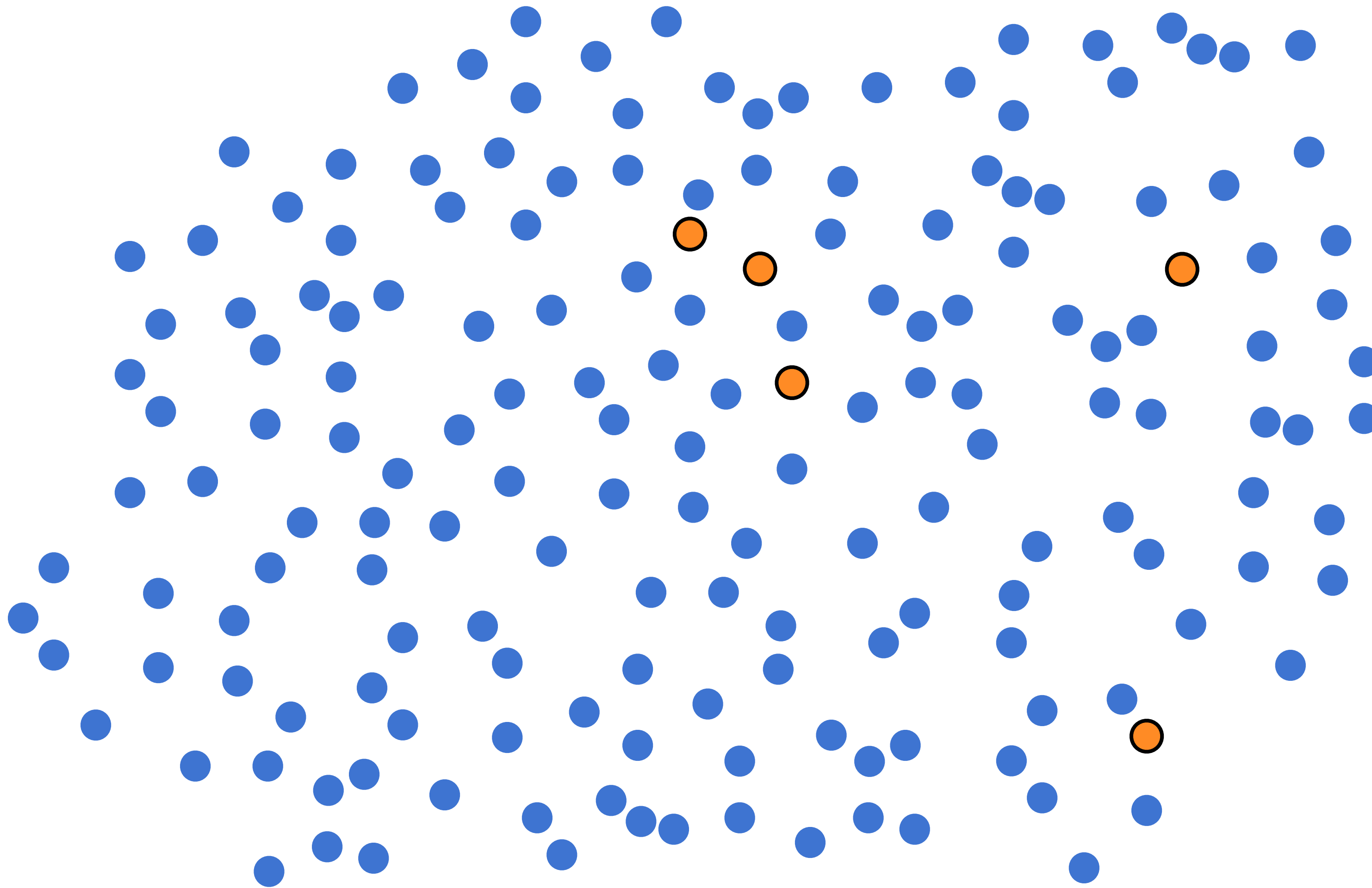
- We can take the average of those temperatures,

$$\frac{99.2 + 98.6 + 99.1 + 97.4 + 98.0}{5}$$

$$\bar{x} = 98.46$$

Sampling Distribution

Of a sample mean, $\bar{X}$



- We can repeat the process again, and get a different result because we randomly selected different people

- $\mu = 98.6$

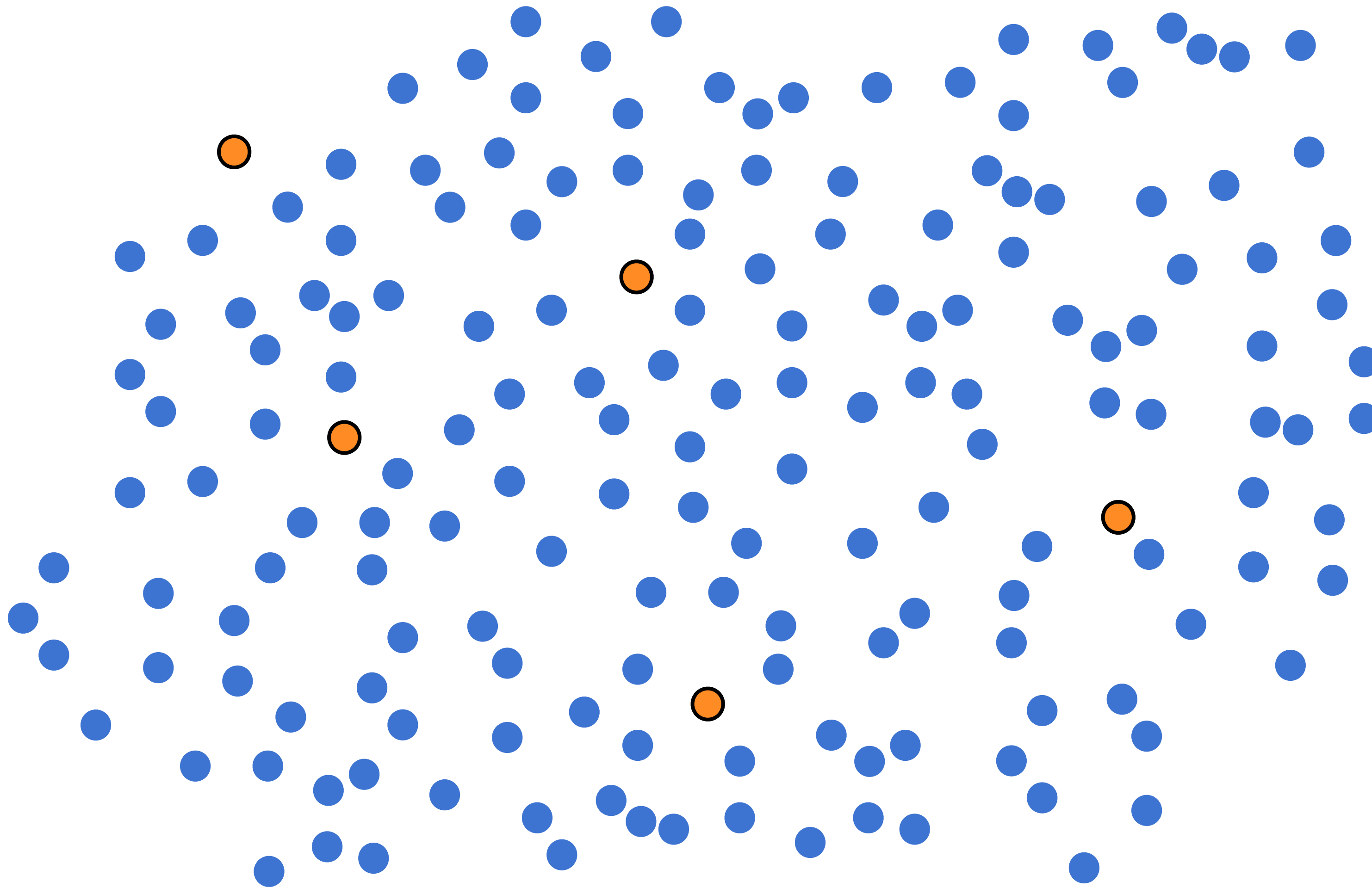
- e.g.,

$$\frac{97.2 + 96.6 + 100.1 + 99.4 + 98.0}{5}$$

$$\bar{x} = 98.26$$

Sampling Distribution

Of a sample mean, $\bar{X}$



- And again...

- $\mu = 98.6$

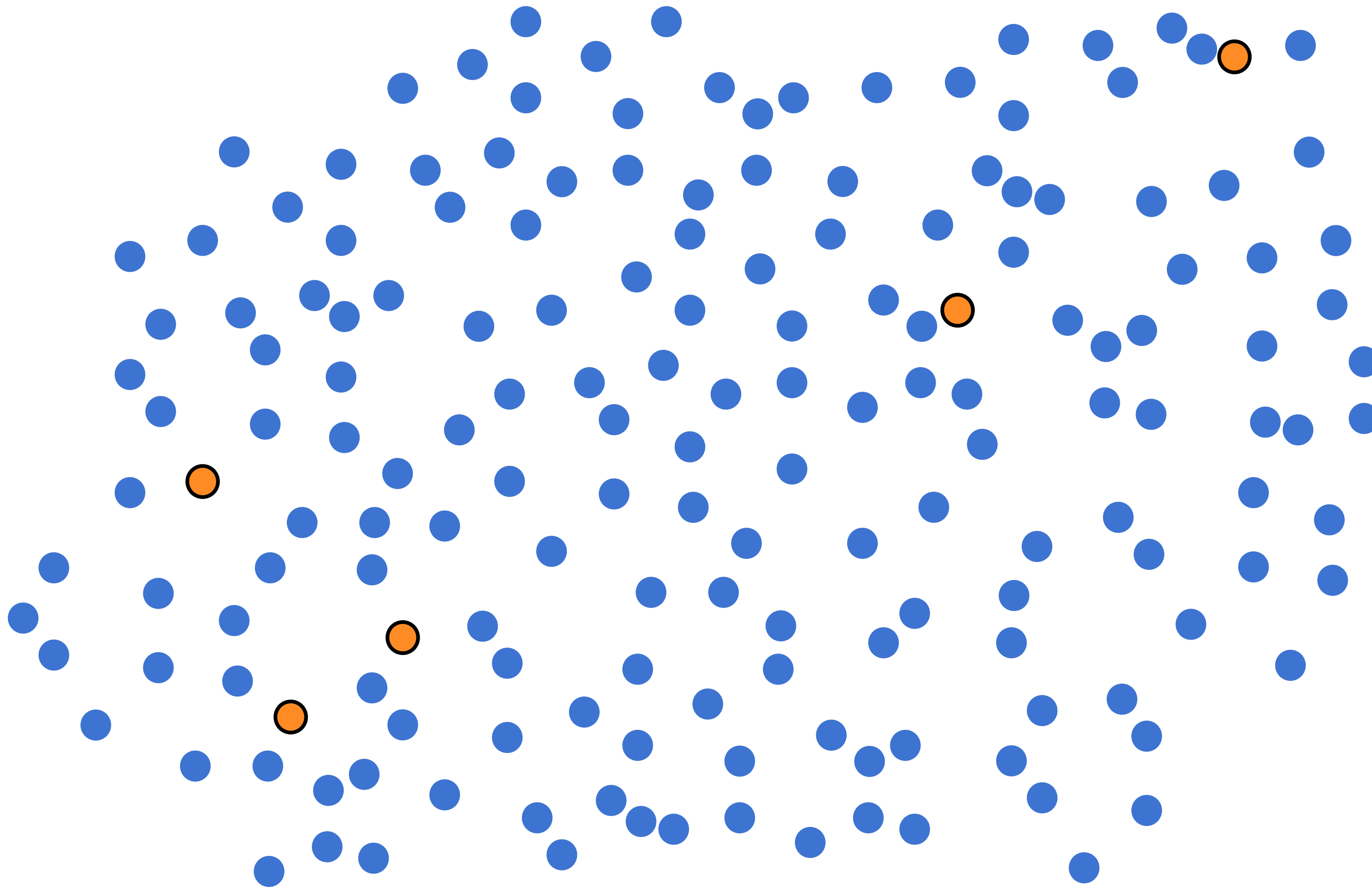
- e.g.,

$$\frac{99.2 + 98.6 + 99.1 + 97.4 + 98.0}{5}$$

$$\bar{x} = 98.46$$

Sampling Distribution

Of a sample mean, $\bar{X}$



- And *again*...

- $\mu = 98.6$

- e.g.,

$$\frac{97.1 + 96.6 + 98.2 + 97.7 + 96.0}{5}$$

$$\bar{x} = 97.12$$

Sampling Distribution

Of a sample mean, $\bar{X}$

- If $\bar{X}$ is a random variable, it must have a probability distribution
 - We call the distribution of $\bar{X}$ the **sampling distribution** of the **sample mean**
 - The idea is that a value computed from a random sample of people also has a probability distribution, because it is random
 - If we know the distribution of the population, we should be able to say something about the distribution of a sample from it

Notation

Sample means

	<u>Mean</u>	<u>Proportion</u>
<u>Population parameter</u> (not random, fixed but unknown value)	μ	
<u>Random variable</u> (random quantity, for hypothetical samples that haven't yet been taken)	$\bar{X}$	
<u>Actual computed value</u> from a sample that <i>has</i> been taken (not random)	$\bar{x}$	

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Sampling Distributions

- In the body temperature example, which of the following best describes the sampling distribution of the sample mean?
 - A. The distribution of body temperatures for all adults in the population
 - B. The distribution of body temperatures for the 5 people in one given sample
 - C. The distribution of possible sample means from many random samples of the same size
 - D. The probability distribution of the population mean μ

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Sampling Distributions

- In the body temperature example, which of the following best describes the sampling distribution of the sample mean?
 - A. The distribution of body temperatures for all adults in the population
 - B. The distribution of body temperatures for the 5 people in one given sample
 - C. The distribution of possible sample means from many random samples of the same size.**
 - D. The probability distribution of the mean μ

Sampling from a Bernoulli Distribution

$$X \sim \text{Bernoulli}(0.85)$$

- Suppose that 85% of U.S. adults have been vaccinated for COVID-19.
 - Note that the number 85% is a population **parameter**, because it is the proportion in the entire population
 - Notation: from here forward we will use p = population proportion (notice that this is the same notation we used for the Bernoulli distribution anyway!)
- Let $X = 1$ if a randomly selected U.S. adult has been vaccinated for COVID, and 0 if not.
- What is the probability that $X = 0$?
 - [PollEv.com/stats8](https://www.pollEv.com/stats8)



Sampling from a Bernoulli Distribution

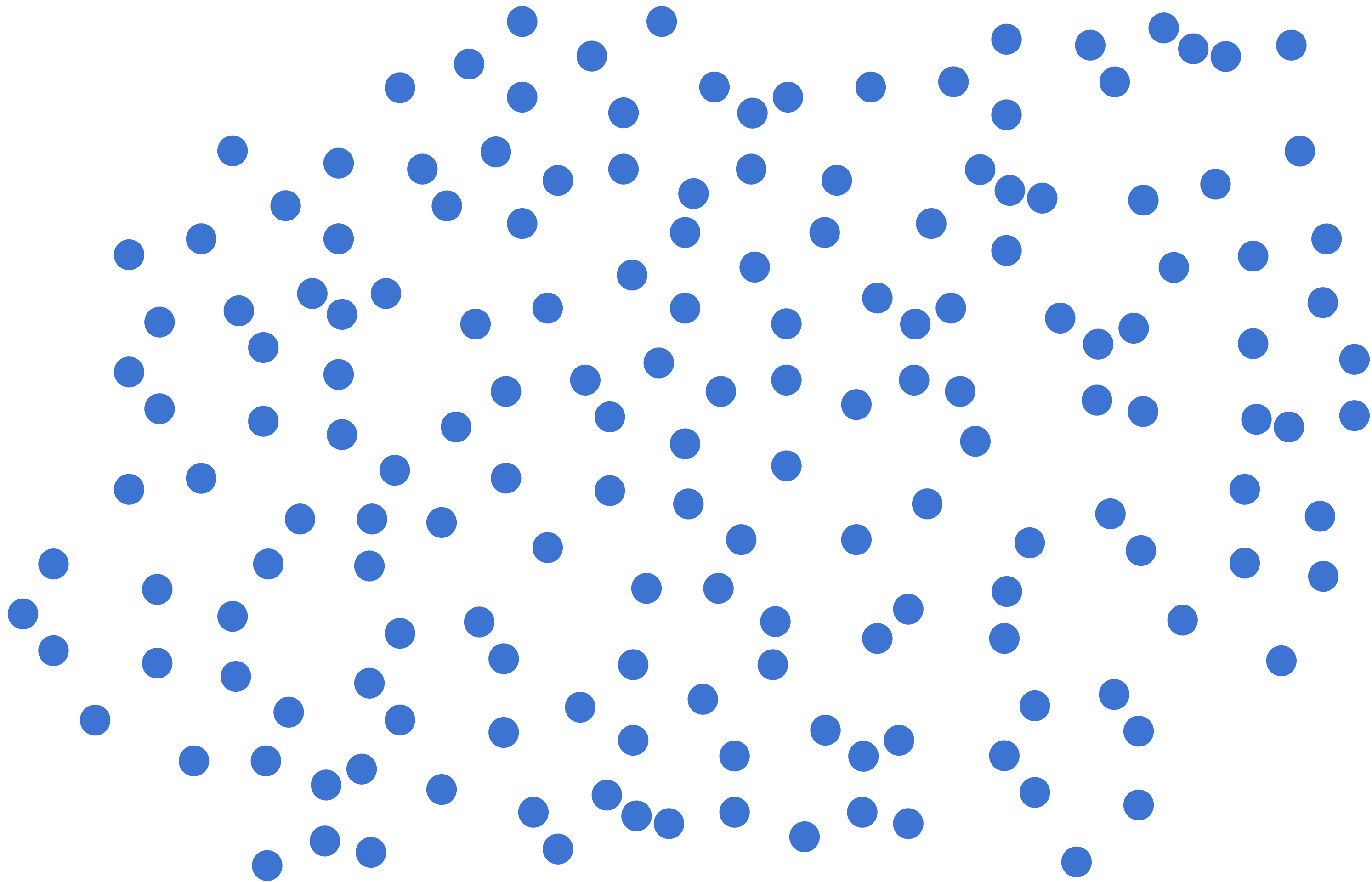
$$X \sim \text{Bernoulli}(0.85)$$

- Suppose that 85% of U.S. adults have been vaccinated for COVID-19.
- Similar to before, we can sample many people and take their average,
 - $X_1 \sim \text{Bernoulli}(0.85), X_2 \sim \text{Bernoulli}(0.85), \dots, X_n \sim \text{Bernoulli}(0.85)$
 - $\frac{X_1 + X_2 + \dots + X_n}{n} = \frac{\# \text{ of sampled 1s}}{n} = \text{sample proportion of 1s}$
- We call this $\hat{P}$
 - here, the proportion of vaccinated individuals in a sample



Sampling Distribution

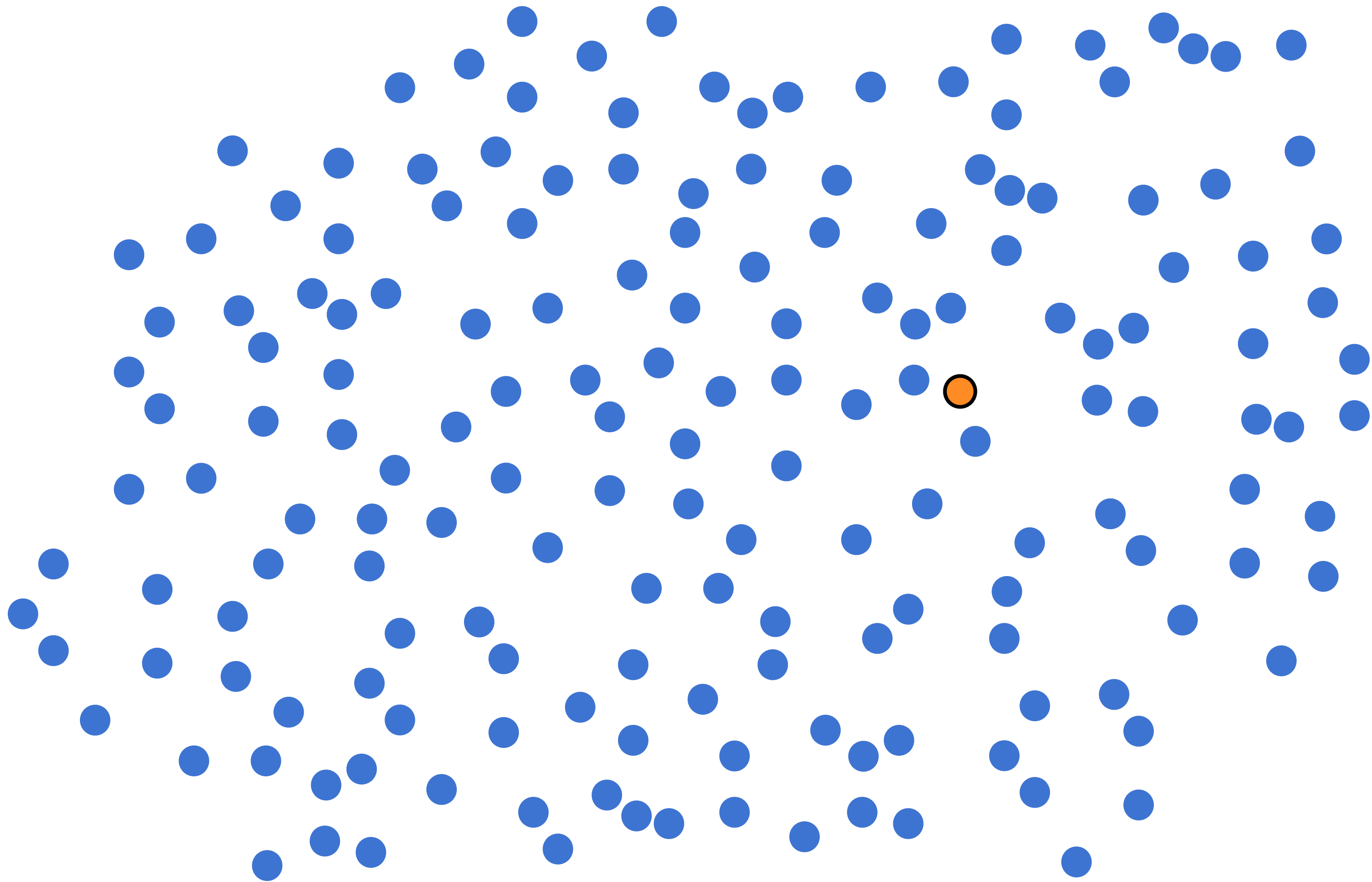
Of a sample proportion, $\hat{p}$



- As with last time, we can imagine we have a population from which we are sampling
- Population proportion of people who have been vaccinated for COVID, $p = 0.85$

Sampling Distribution

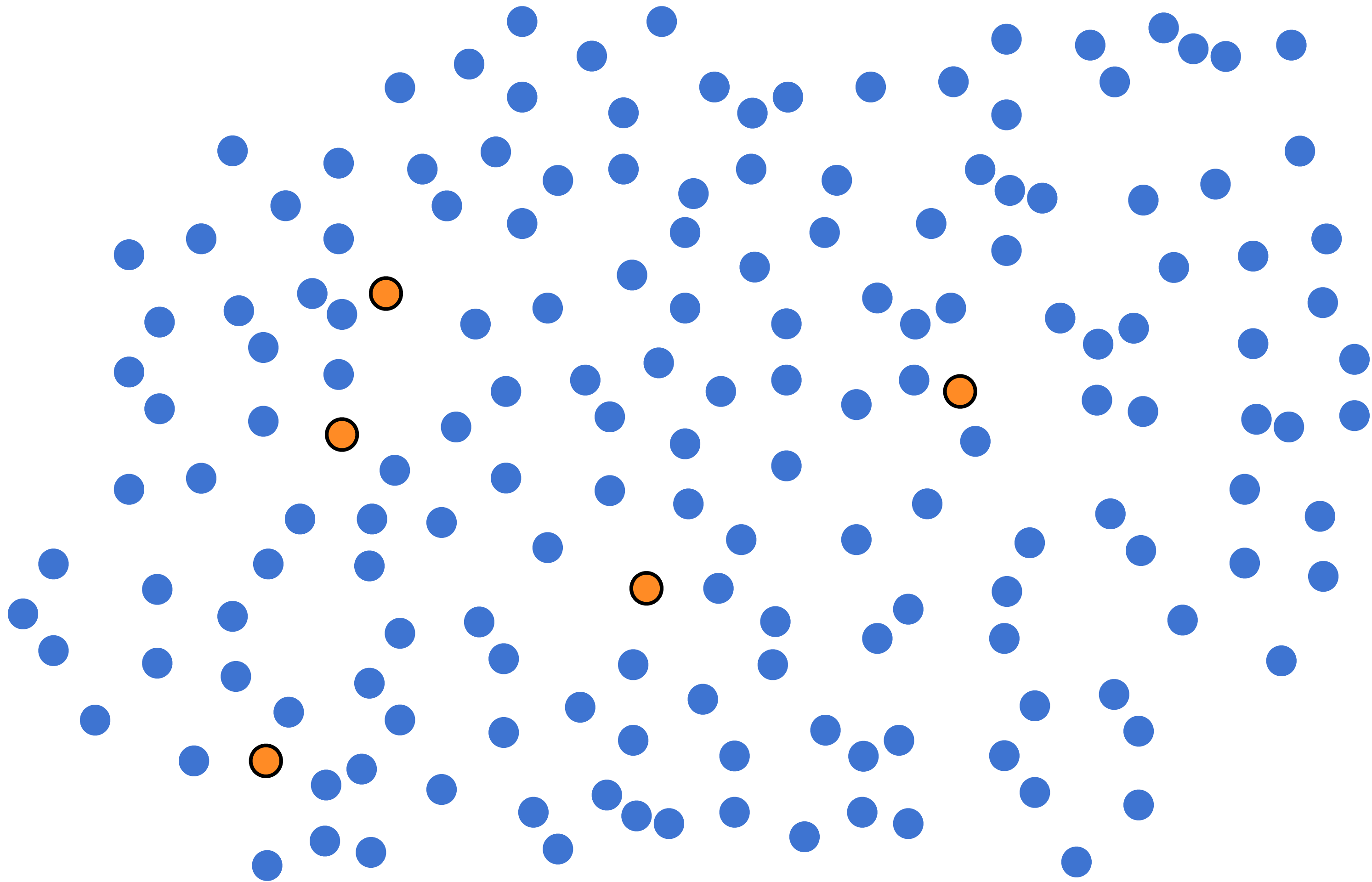
Of a sample proportion, $\hat{p}$



- We can sample a person randomly and check to see whether they are vaccinated for COVID, $p = 0.85$
- e.g., yes (1)

Sampling Distribution

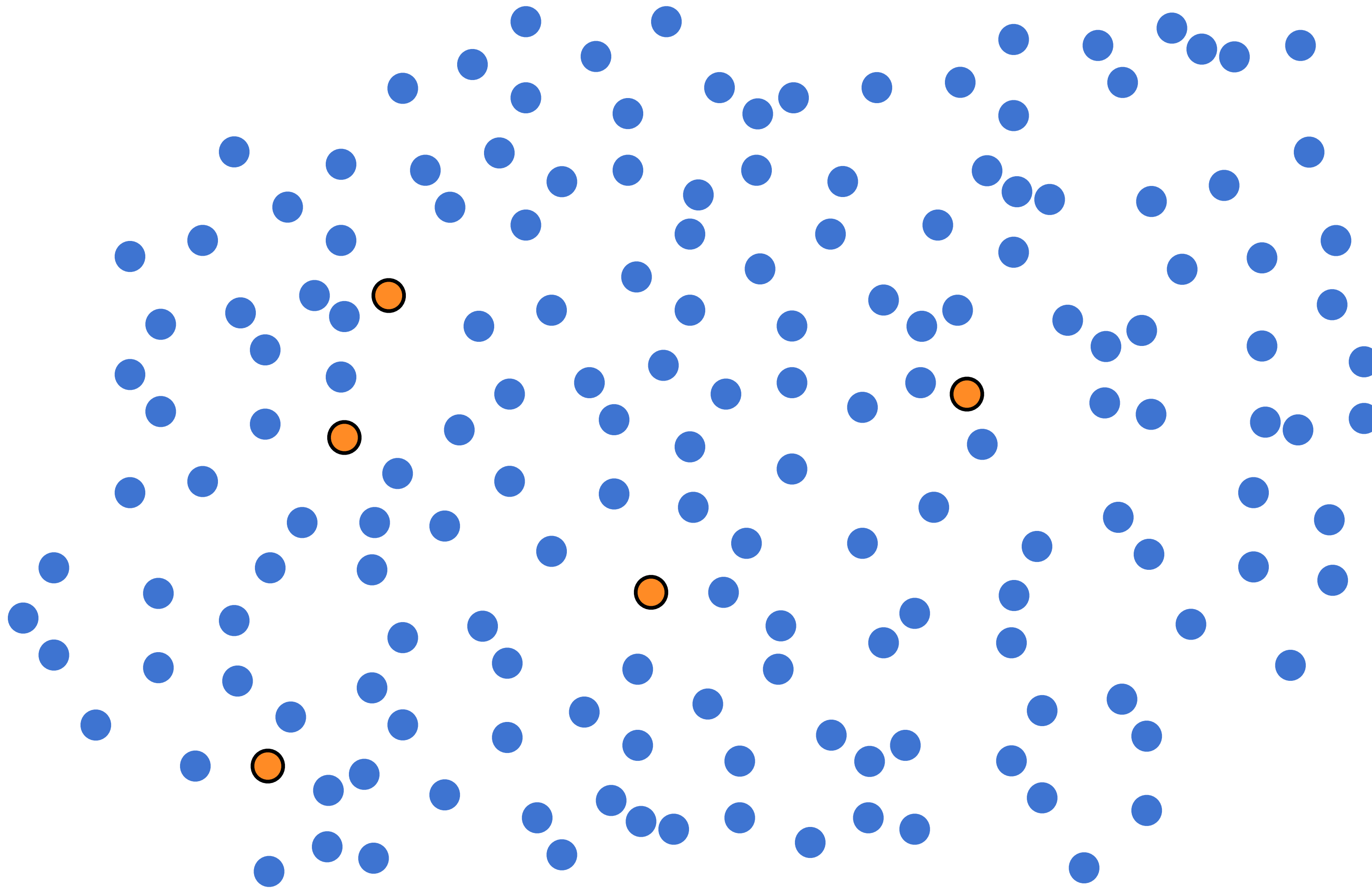
Of a sample proportion, $\hat{p}$



- We can also randomly sample many people, and check whether they are vaccinated for COVID, $p = 0.85$
- e.g., yes (1), no (0), yes (1), yes (1), yes (1)

Sampling Distribution

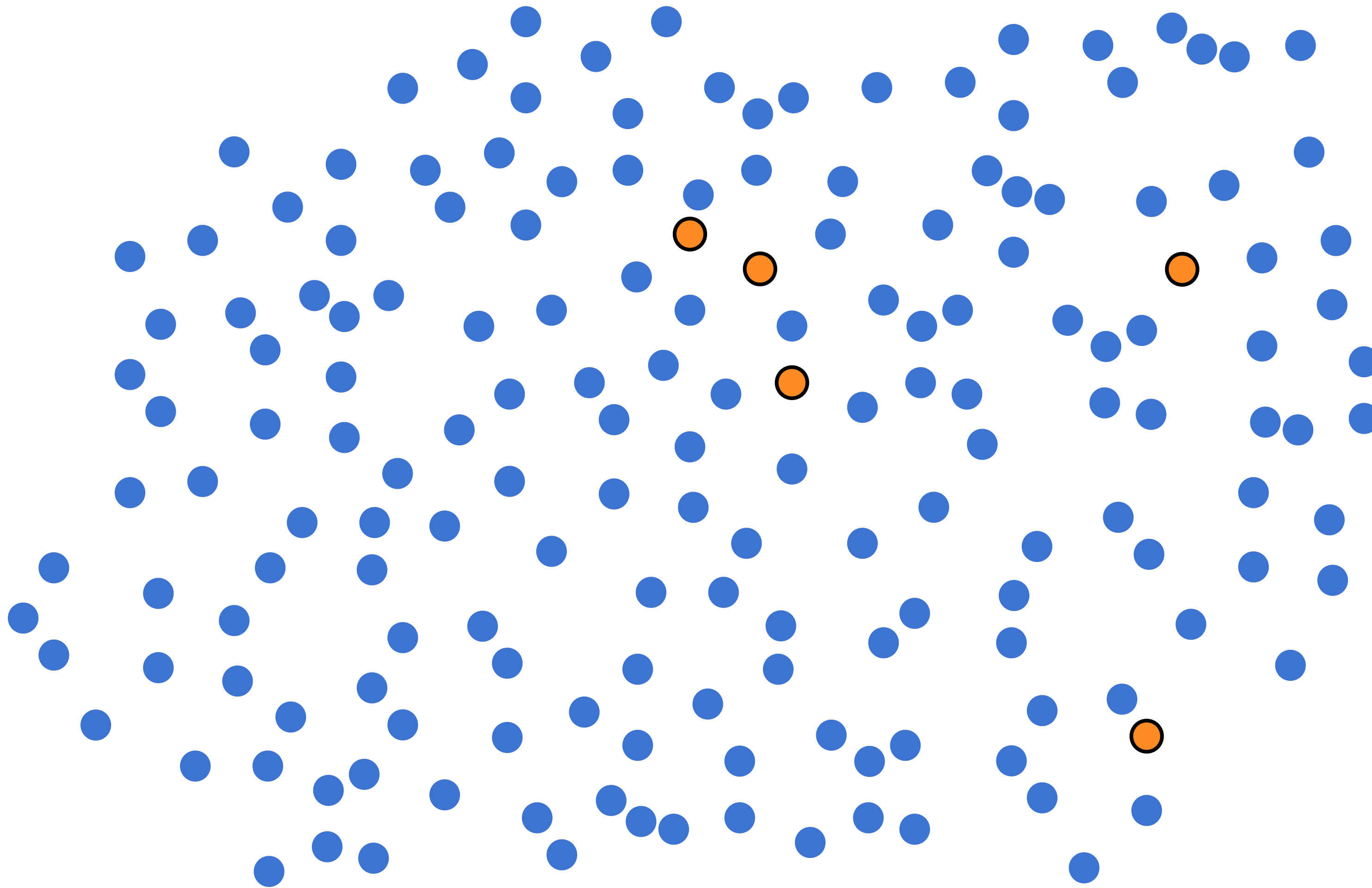
Of a sample proportion, $\hat{p}$



- We can also randomly sample many people, and check whether they are vaccinated for COVID
 - $p = 0.85$
 - e.g., yes (1), no (0), yes (1), yes (1), yes (1)
- We can then get the proportion of people who are vaccinated in this sample,
 - $\hat{p} = (1+0+1+1+1) / 5 = 0.8$

Sampling Distribution

Of a sample proportion, $\hat{p}$



- We can repeat the process again, and get a different result because we randomly selected different people

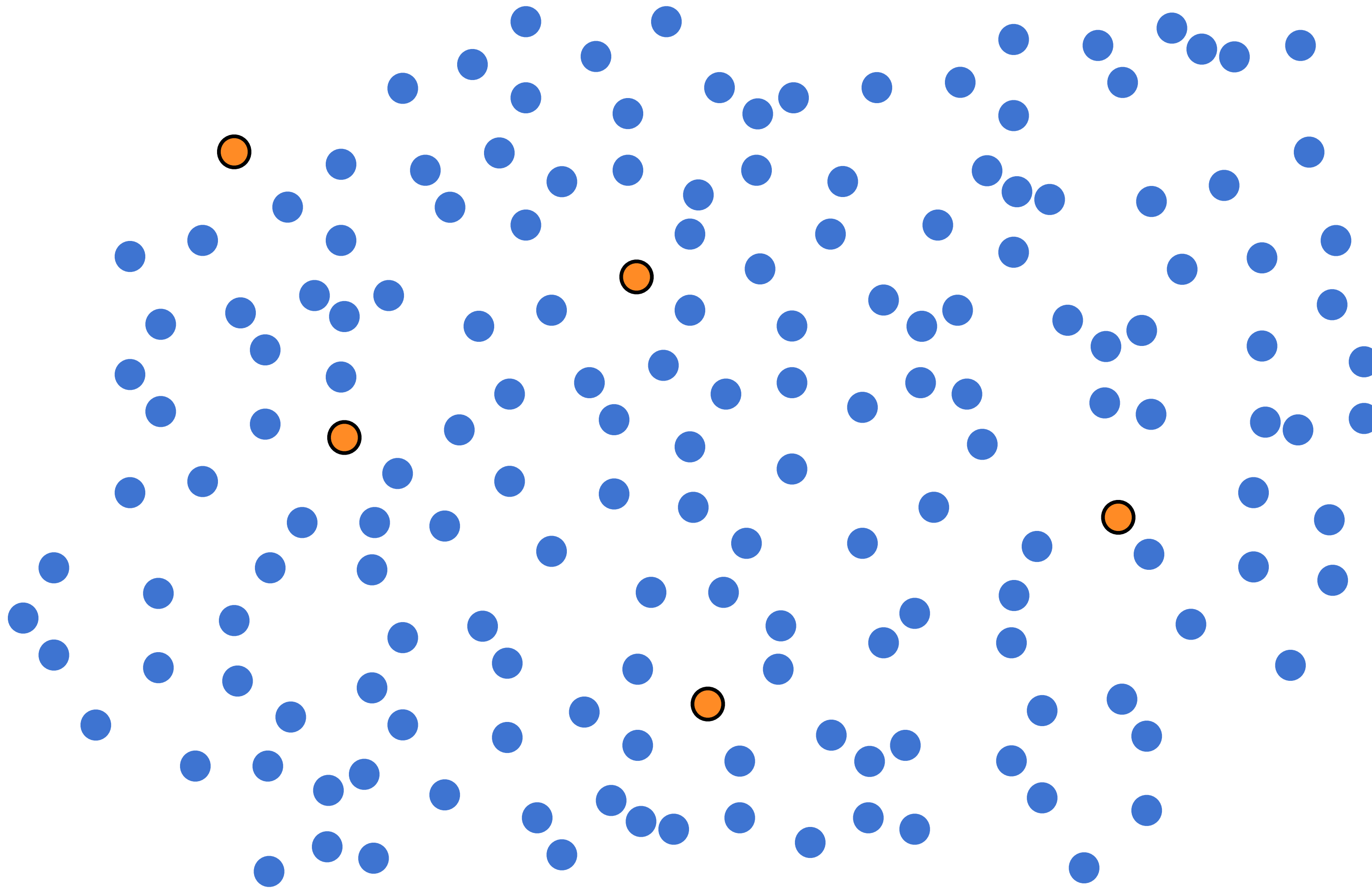
- $p = 0.85$

- e.g.,

$$\hat{p} = \frac{0 + 1 + 1 + 0 + 1}{5} = 0.6$$

Sampling Distribution

Of a sample proportion, $\hat{p}$



- And so on...

- $p = 0.85$

- e.g.,

$$\hat{p} = \frac{1 + 1 + 1 + 1 + 1}{5} = 1.0$$

Notation

Sample means & sample proportions

	<u>Mean</u>	<u>Proportion</u>
<u>Population parameter</u> (not random, fixed but unknown value)	μ	p
<u>Random variable</u> (random quantity, for hypothetical samples that haven't yet been taken)	$\bar{X}$	$\hat{P}$
Actual <u>computed value</u> from a sample that <i>has</i> been taken (not random)	$\bar{x}$	$\hat{p}$

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Notation

- A researcher samples 100 students, interested in what proportion of students sleep at least 7 hours each night. From their survey, they find that 63% report sleeping at least 7 hours each night. Which symbol should represent 0.63?

A. $\bar{x}$

B. $\bar{X}$

C. μ

D. $\hat{p}$

E. $\hat{P}$

F. p

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Notation

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A. $\bar{x}$

B. $\bar{X}$

C. μ

D. $\hat{p}$

E. $\hat{P}$

F. p

Sampling Distributions

Critical Distinctions in Terminology

- Note some similar-sounding but very different terms:
 - Population distribution of the quantitative variable X : refers to the probability distribution of a random variable (the distribution of the value of the variable for all the individuals in the population). (**Mathematical model!**)
 - Sample distribution, or distribution of the sample: describing a histogram/distribution of *one* random sample of n data points (see Topic 2). (**Actual data!**)
 - Sampling distribution of the sample mean $\bar{X}$ or the sample proportion $\hat{p}$: refers to the theoretical probability distribution of the statistic, of *all possible samples* of size n from a given population. (**Mathematical model!**)

Sampling Distributions

Of a statistic

- Statistics are values computed from samples, and because *who is included in the sample* is random, the statistics are also random
- This is the bridge from probability to statistics:
 - Probability: if we know the population, what samples might we get?
 - Statistics: if we only have a sample, what can we say about the population?

★ **Core idea of statistics**: When the sample is randomly selected from the population of interest, the value of the sample statistic should be pretty close to the value of the population parameter

- Intuitive. But:
 1. Can we prove that it's true, mathematically?
 2. How close is “pretty close”? Can we quantify that?

i.i.d. Random Variables

- **i.i.d.** is a common term in statistics that stands for:

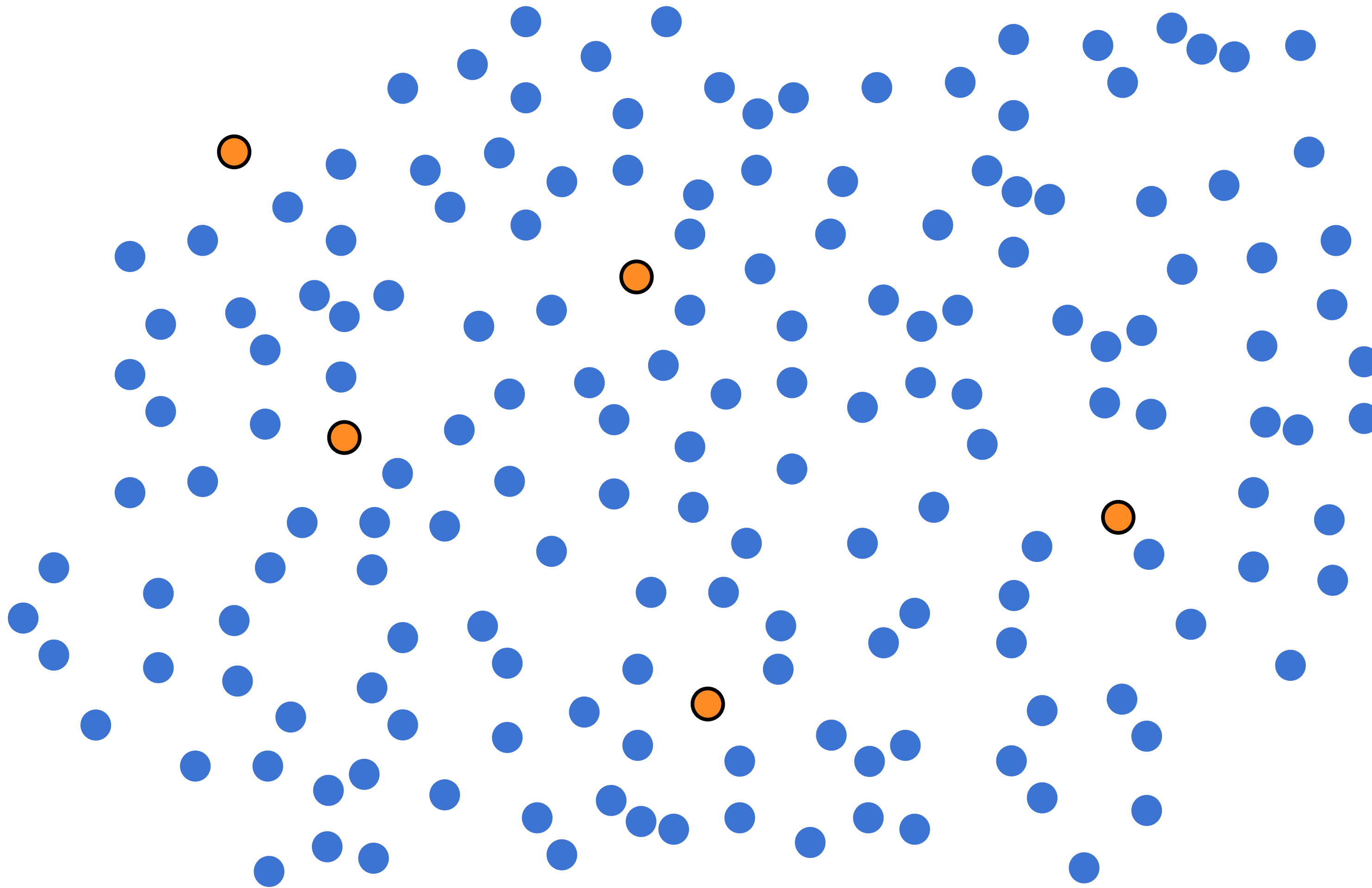
- **I**ndependent, and
- **I**dentically **d**istributed.

- We will often denote this:

$$X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} [\text{Some probability distribution}]$$

- Independent means that the random variables do not “affect” each other; their probability distributions don’t depend on one another
- Identically distributed means they all have the exact same probability distribution

i.i.d. Random Variables



- If we take a **random sample** from a population, we should have **i.i.d.** random variables
- Independent (or close enough to independent) because the sample is randomly selected from a large population
- Identically distributed because they all have the population distribution

i.i.d. Random Variables

$X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim}$ [Some probability distribution with mean = μ and variance = σ^2]

- The property of being i.i.d. is nice because it gives us some interesting properties regarding the mean and variance
 - (showing this is beyond the scope of this class, but come to office hours if curious)

1. Mean/expected value: $E \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) = \mu$

2. Variance: $\text{Var} \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) = \frac{\sigma^2}{n}$

i.i.d. Random Variables

$X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim}$ [Some probability distribution with mean = μ and variance = σ^2]

- The property of being i.i.d. is nice because it gives us some interesting properties regarding the mean and variance
 - (showing this is beyond the scope of this class, but come to office hours if curious)
- 1. Mean/expected value: the expected value of the mean of the random variables is equal to the mean of any one of the random variables (they should all be the same)
- 2. Variance: the variance of the mean of the random variables is equal to $1/n$ times the variance of any one of the random variables (they should all be the same)

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Sampling distribution of $\bar{X}$

- Suppose the population mean systolic blood pressure among healthy adults is $\mu = 120$, with population standard deviation $\sigma = 15$. A researcher plans to randomly sample $n = 25$ healthy adults and compute the sample average. What is the mean of the sampling distribution of the sample mean, $\bar{X}$?

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Sampling distribution of $\bar{X}$

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- 120

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Sampling distribution of $\bar{X}$

- Suppose the population mean systolic blood pressure among healthy adults is $\mu = 120$, with population standard deviation $\sigma = 15$. A researcher plans to randomly sample $n = 25$ healthy adults and compute the sample average. What is the **standard deviation** of the sampling distribution of the sample mean, $\bar{X}$?

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Sampling distribution of $\bar{X}$

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Sampling distribution of $\bar{X}$

- Suppose the population mean systolic blood pressure among healthy adults is $\mu = 120$, with population standard deviation $\sigma = 15$. A researcher plans to randomly sample $n = 25$ healthy adults and compute the sample average. If the sample size increased from 25 to 100, what would happen to the standard deviation of the sample mean, $\bar{X}$?
 - A. It would get 4 times larger
 - B. It would get 4 times smaller (i.e., be $1/4$ the size)
 - C. It would get twice as large
 - D. It would get twice as small (i.e., be $1/2$ the size)
 - E. It would not change
 - F. Something else

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 - D. It would get twice as small (i.e., be $1/2$ the size)**
 - E. It would not change
 - F. Something else

★The Central Limit Theorem★

- We talked about the mean and variance of a sample mean, but what about the whole distribution?
- If:

1. The random variables are i.i.d., and
2. The sample size is “large” (more later), then:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \sim N(\mu, \sigma^2/n)$$

- When the sample size becomes very big, the sampling distribution of the sample average becomes approximately **normal** (exact in the limit $n \rightarrow \infty$)

★The Central Limit Theorem★

- If:
 1. The random variables are i.i.d., and
 2. The sample size is “large” (more later), then:
$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \dot{\sim} N(\mu, \sigma^2/n)$$
- For large n , the sampling distribution of $\bar{X}$ is approximately normal (exact in the limit $n \rightarrow \infty$)
- **Again: for any* population distribution, if you take a large random sample from it and take the average, that sample average has an approximately normal sampling distribution!**
- Simulate it for yourself

* there are some mild conditions on this but they would be in very weird situations, not in this class

The Central Limit Theorem (CLT)

- If:
 1. The random variables are i.i.d., and
 2. The sample size is “large” (more later), then:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \dot{\sim} N(\mu, \sigma^2/n)$$

- What about $\hat{P}$?
 - Remember that a sample proportion is a specific type of sample mean!
 - CLT still applies
 - If $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$, where n is large, then

$$\hat{P} = \frac{X_1 + X_2 + \dots + X_n}{n} \dot{\sim} N\left(\text{[pop. mean]}, \frac{1}{n}[\text{pop. variance}]\right)$$

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CLT for $\hat{P}$

- Why is a sample proportion a type of sample mean?
 - A. Because it is normally distributed.
 - B. Because it averages 0s and 1s.
 - C. Because it equals the population proportion.
 - D. Because proportions are not random variables.

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CLT for $\hat{P}$

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 - A. Because it is normally distributed.
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The Central Limit Theorem (CLT)

- What about $\hat{P}$?
 - CLT still applies
 - If $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$, where n is large, then

$$\hat{P} = \frac{X_1 + X_2 + \dots + X_n}{n} \stackrel{\sim}{\sim} N \left([\text{pop. mean}], \frac{1}{n} [\text{pop. variance}] \right)$$

- What are the population mean and variance?
 - Remember for Bernoulli:
 - $E(X) = p$, $\text{Var}(X) = p(1 - p)$
 - So:

$$\hat{P} = \frac{X_1 + X_2 + \dots + X_n}{n} \stackrel{\sim}{\sim} N \left(p, \frac{p(1 - p)}{n} \right)$$

Conditions for CLT

a.k.a., we keep saying “if n is large” — how large?

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \quad \hat{P} \dot{\sim} N\left(p, \frac{p(1-p)}{n}\right)$$

The math requires certain conditions for the statements above using CLT to apply.

1. i.i.d. sample — sampled individuals are unrelated but come from the same population distribution

✓ Satisfied when we use probability sampling from a large population

2. Large enough sample:

• For $\bar{X}$:

- $n = 10$ is sufficient if we think the population distribution is “nice” — symmetric, without tons of outliers
- $n = 20$ is generally needed for distributions with more skew
- $n = 30$ is good enough for almost all distributions

• For $\hat{P}$:

- Rule of thumb is that $np \geq 10$ and $n(1-p) \geq 10$ is sufficient for CLT to “kick in”
- In practice, check this by subbing $\hat{p}$ in for p
- Basically: do we have enough sampled people in the “yes” group and in the “no” group?

Conditions for CLT

a.k.a., we keep saying “if n is large” — how large?

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \bigg| \quad \hat{P} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

The math requires certain conditions to apply.

1. i.i.d. sample — sample must come from a population distribution

✓ Satisfied when we observe a random sample

2. Large enough sample size

• For $\bar{X}$:

- $n = 10$ is sufficient if we think the population distribution is “nice” — symmetric, without tons of outliers
- $n = 20$ is generally needed for distributions with more skew
- $n = 30$ is good enough for almost all distributions

 **on an exam, it will never be ambiguous whether it applies or not! but, it can be ambiguous in real life, so keep that in mind.** 

- Rule of thumb is that $np \geq 10$ and $n(1-p) \geq 10$ is sufficient for CLT to “kick in”
- In practice, check this by subbing $\hat{p}$ in for p
- Basically: do we have enough sampled people in the “yes” group and in the “no” group?

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CLT for $\hat{P}$

- A researcher takes a random sample of $n = 20$ people. In the sample, $\hat{p} = 0.95$. They want to use a normal approximation for the sample proportion. What issue should they notice?

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CLT for $\hat{P}$

- A researcher takes a random sample of $n = 20$ people. In the sample, $\hat{p} = 0.95$. They want to use a normal approximation for the sample proportion. Which issue should they notice?
- $n(1 - \hat{p}) = 20(0.05) = 1$, we don't have enough people in the "no" group for CLT to apply

Connection to Inference

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CLT

- The Central Limit Theorem says that, for a large random sample, the sampling distribution of the sample mean is approximately normal. What is especially useful about this?
 - A. It means the original population distribution must be normal.
 - B. It means each individual value in the sample is approximately normal.
 - C. It gives us a way to model how sample means vary from sample to sample.
 - D. It means the sample mean will usually equal the population mean.

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CLT

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 - C. It gives us a way to model how sample means vary from sample to sample.**
 - D. It means the sample mean will usually equal the population mean.

Why does CLT matter?

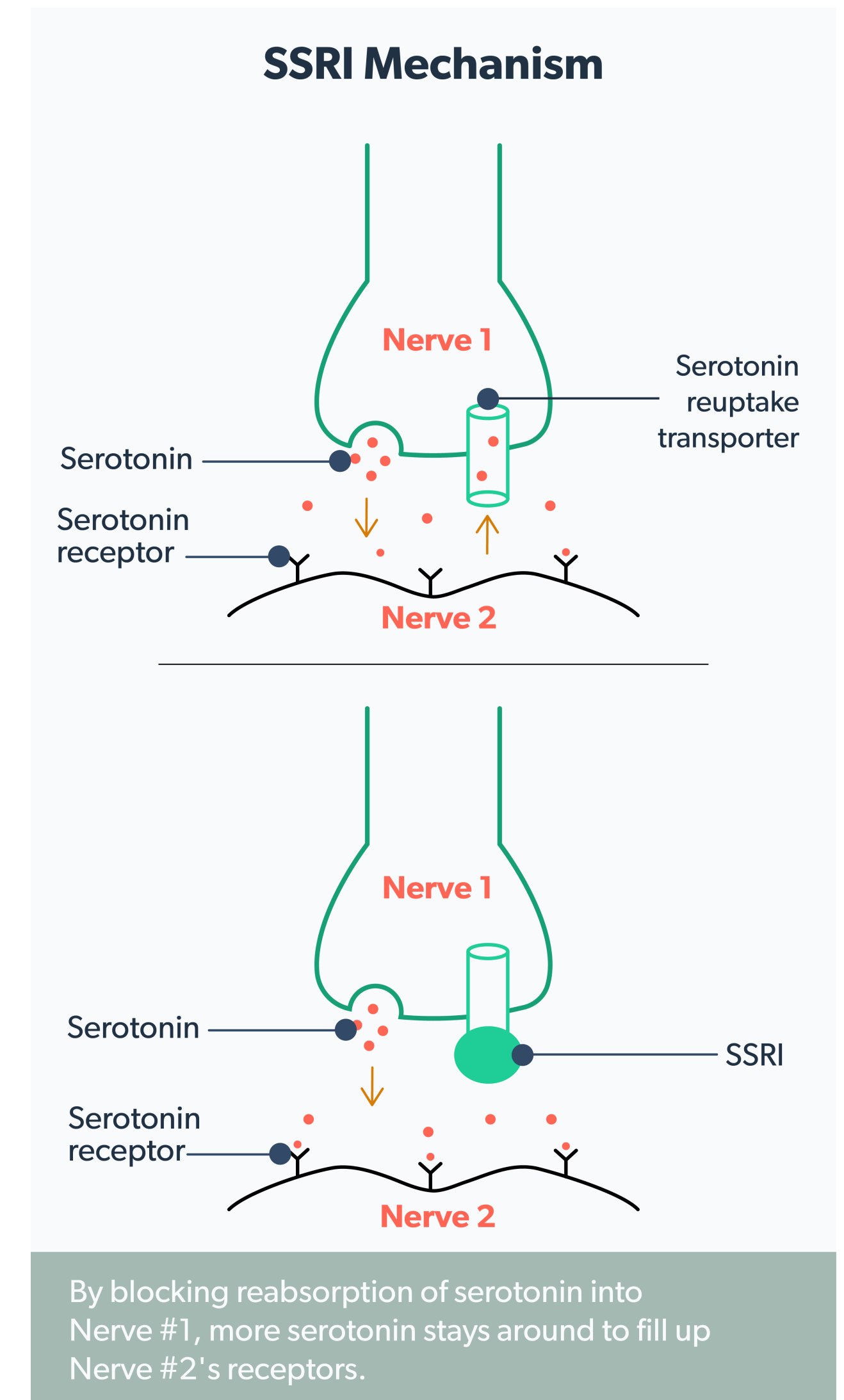
- The CLT is useful because it gives us a model for how much sample statistics vary from sample to sample (this variation is exactly what “uncertainty” means)
- Let’s say we don’t know anything about a population distribution, but we want to know its mean (let the mean and variance of the population distribution be μ and σ^2 , respectively)
 - What’s the average height of 10-year-old children in the U.S.?
 - What’s the average blood pressure of healthy adults?
 - How much do symptoms of a disease change on average when people take a certain medication?
- **Regardless** of the distribution of the population (which we basically never know), if we take a large enough random sample, we know that

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- So... can we work backwards from the sample to learn about μ ? (Yes!)

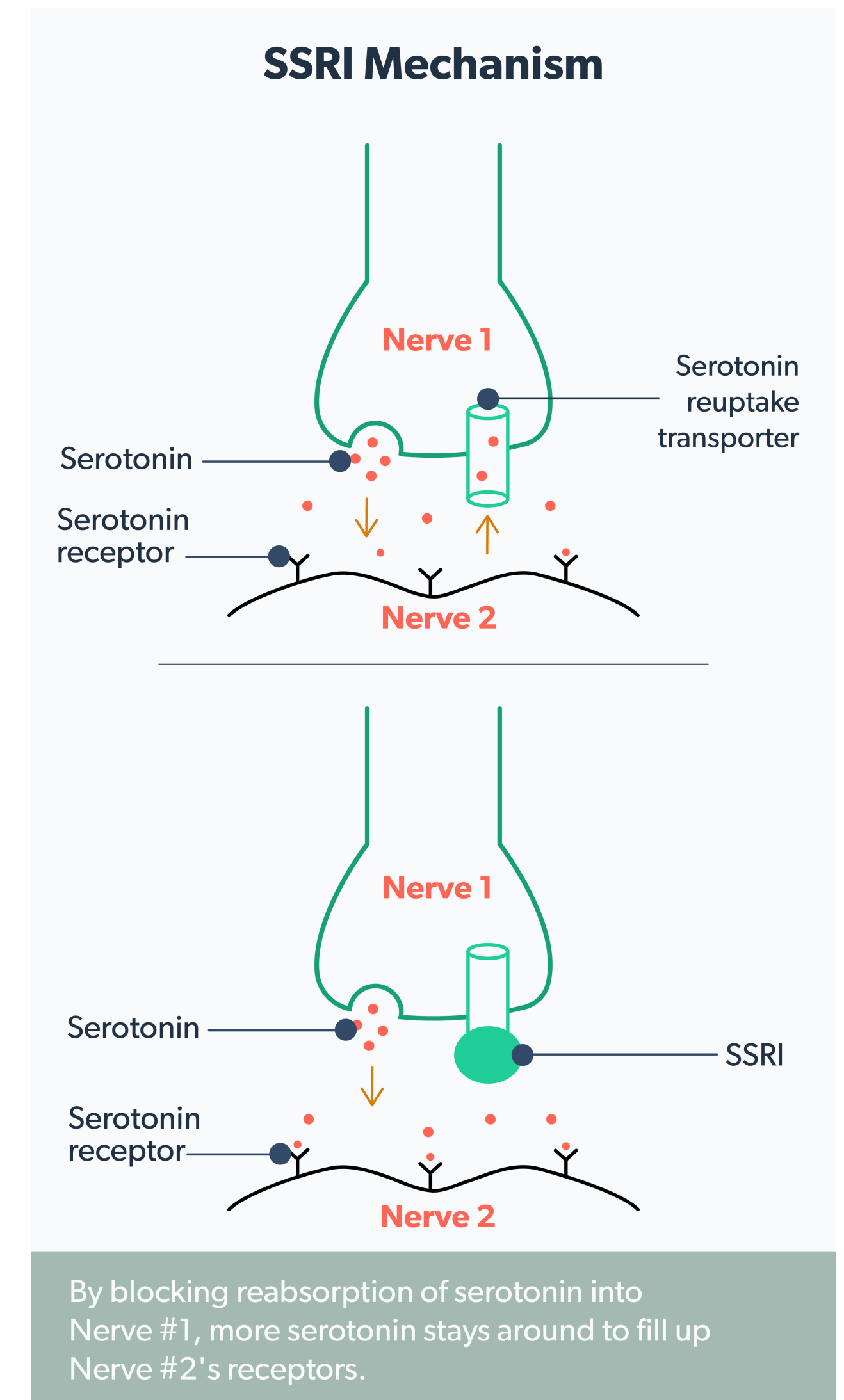
Example: Symptom Improvement

- Let's say we're interested in how much a person with depression's symptoms change after starting a certain SSRI
- The population we are interested in is the population of adults in the U.S. with depression
- We can score them on a depression scale based on a survey before and after starting an SSRI (this is a real thing: Patient Health Questionnaire-9, or PHQ-9, which scores depression from 0 to 27)
- Let X be the change in PHQ-9 score after 8 weeks for a randomly selected adult who starts this SSRI



Example: Symptom Improvement

- Let's say we're interested in how much a person with depression's symptoms change after starting a certain SSRI
- Let X be the change in PHQ-9 score after 8 weeks for a randomly selected adult who starts this SSRI
- ☆ What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?
 - Target: want to learn about μ , the average in the population of interest, using $\bar{X}$, the average in our sample



Example: Symptom Improvement

- Let X = the change in PHQ-9 score after 8 weeks for a randomly selected adult who starts this SSRI
- What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?
 - Target: want to learn about μ
- We don't know anything about the distribution of X . But we know that for a hypothetical random sample from the population of U.S. adults with depression:

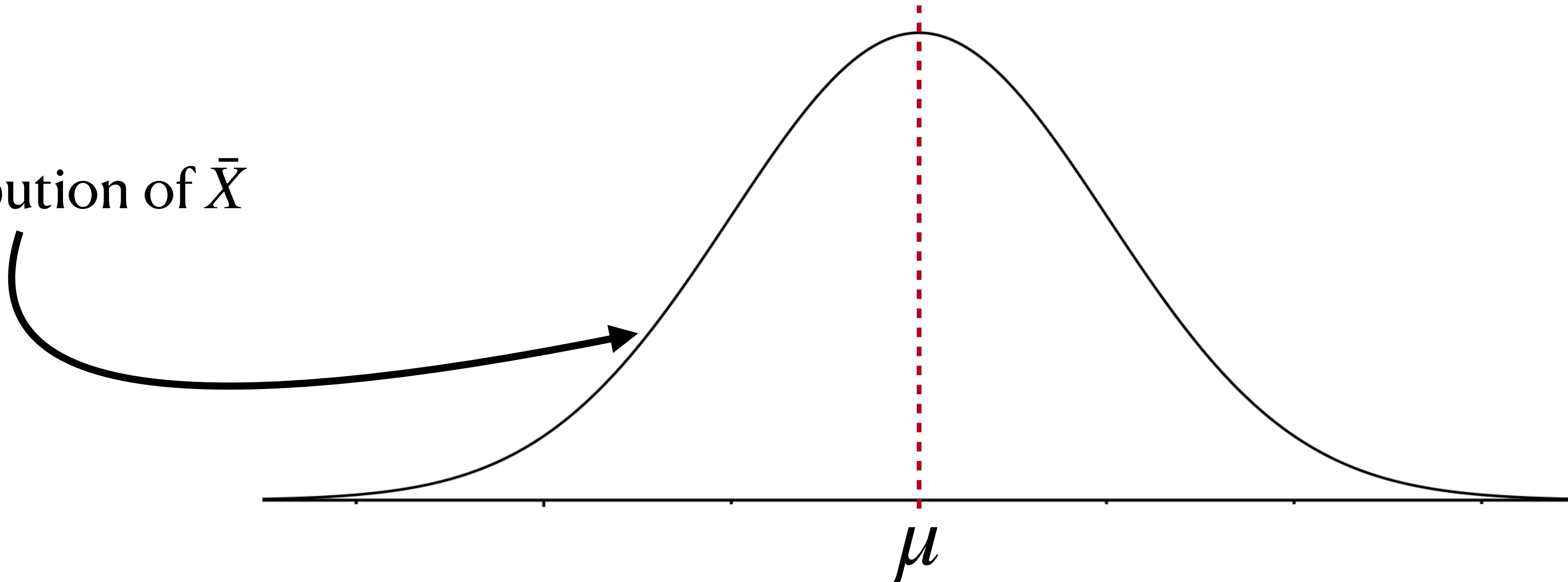
$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Example: Symptom Improvement

- ☆ What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

Distribution of $\bar{X}$

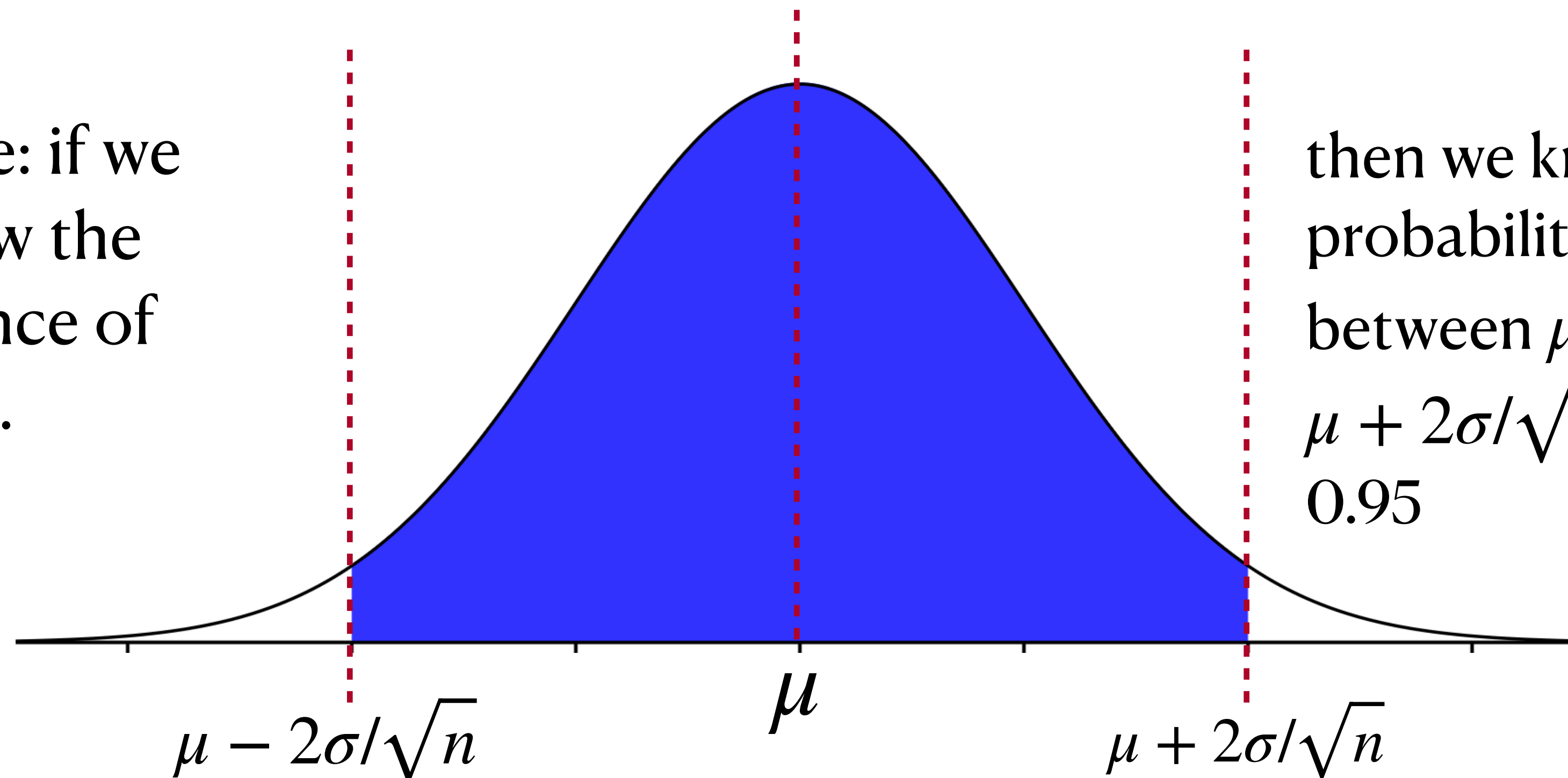


Example: Symptom Improvement

- ☆ What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

By empirical rule: if we pretend we know the mean and variance of the population...



then we know that the probability our sample mean is between $\mu - 2\sigma/\sqrt{n}$ and $\mu + 2\sigma/\sqrt{n}$ is about 95% or 0.95

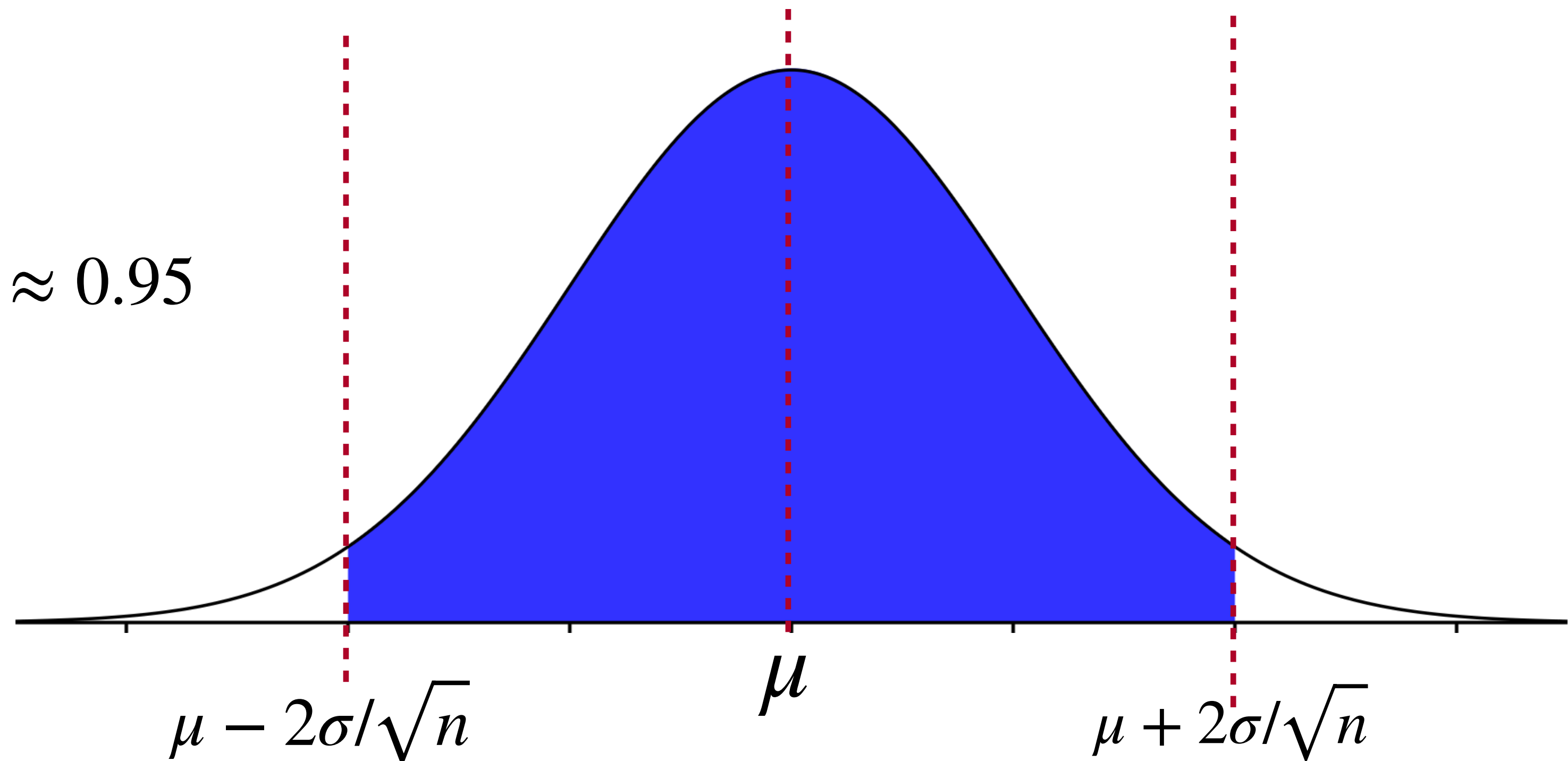
Example: Symptom Improvement

- ☆ What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

In other words,

$$P\left(\mu - 2\sigma/\sqrt{n} \leq \bar{X} \leq \mu + 2\sigma/\sqrt{n}\right) \approx 0.95$$



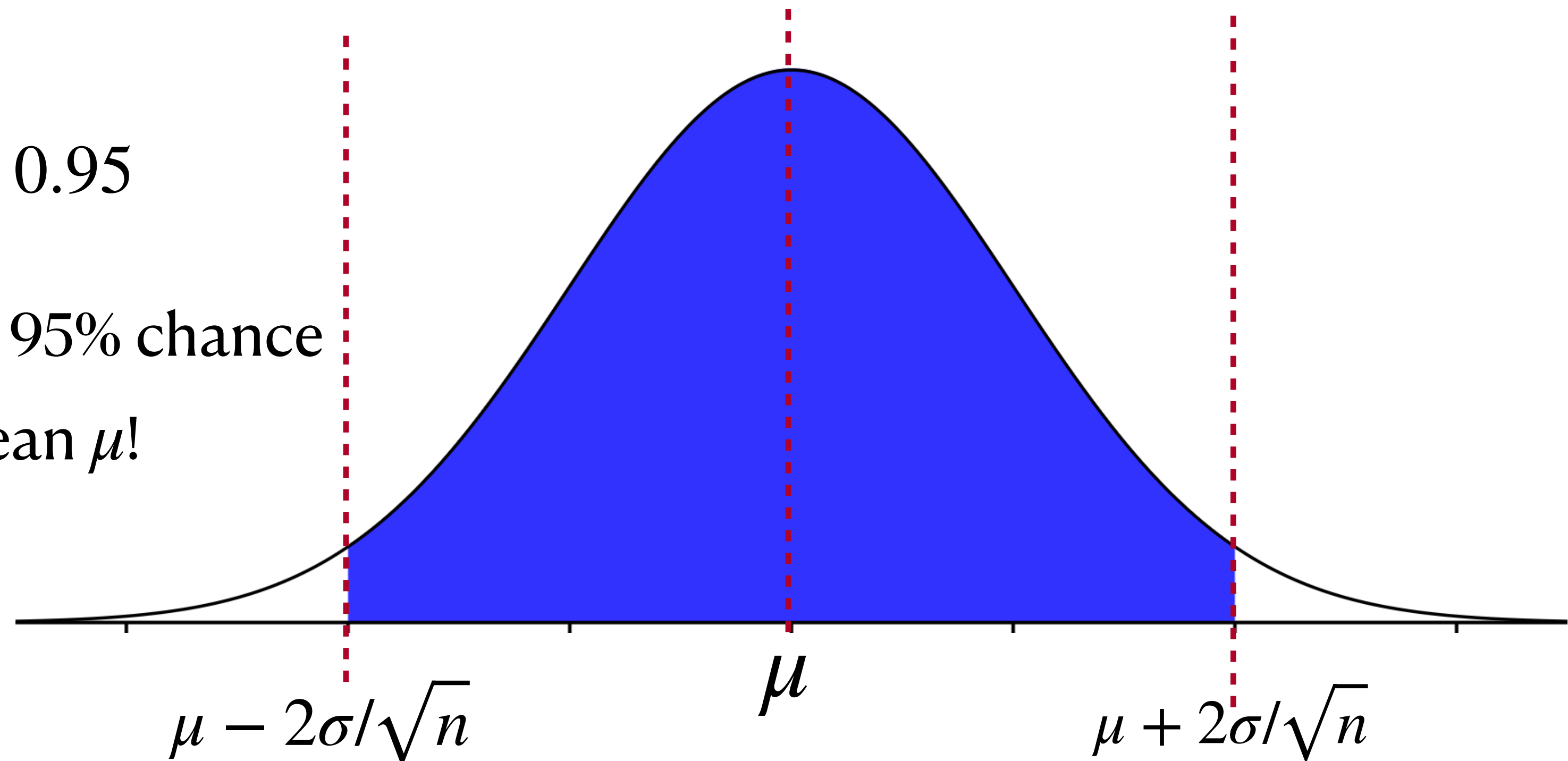
Example: Symptom Improvement

- ☆ What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

$$P\left(\bar{X} - 2\sigma/\sqrt{n} \leq \mu \leq \bar{X} + 2\sigma/\sqrt{n}\right) \approx 0.95$$

The random interval $\bar{X} \pm 2\sigma/\sqrt{n}$ has a 95% chance of containing the (fixed) population mean μ !



Example: Symptom Improvement

- ☆ What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

- Motivates two (related) types of statistical inference:

Hypothesis test: make an assumption about μ based on a hypothesis, then find the probability of getting your observed sample mean if your assumption is true

→if the probability is really low, maybe the assumption was incorrect!

Confidence interval: method for developing an interval of values based on the sample mean $\bar{X}$ that should include μ with some pre-selected probability (often 0.95)

Example: Symptom Improvement

Hypothesis Test Intuition

☆ Do (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

- Knowing that:

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

- Let's say we assume that $\mu = 0$ (the **null hypothesis** — hypothesis of “no difference”)
 - Using that information, we can compute probabilities of $\bar{X}$, the sample mean, using the normal distribution
 - If the probabilities we compute are really small, we might reject our original assumption and say the population mean is not zero

Example: Symptom Improvement

Hypothesis Test Intuition

- ☆ Do (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

We will have to estimate $\sigma/\sqrt{n}$, because σ is not known — this estimate is called the **standard error** of the sample mean

- Knowing that:

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

(the standard deviation of the random variable $\bar{X}$ is the theoretical value $\sigma/\sqrt{n}$, which we don't know)

- Let's say we assume that $\mu = 0$ (the **null hypothesis** — hypothesis of “no difference”)
 - Using that information, we can compute probabilities of $\bar{X}$, the sample mean, using the normal distribution
 - If the probabilities we compute are really small, we might reject our original assumption and say the population mean is not zero

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- Suppose we assume $\mu = 0$, meaning the SSRI has no average symptom improvement in the population. Then we observe a sample mean improvement that would be extremely unlikely if $\mu = 0$. What is the basic intuition of a hypothesis test here?
 - A. The sample was collected incorrectly
 - B. The assumption $\mu = 0$ may not be believable
 - C. The population mean actually is 0
 - D. The CLT cannot apply

Example: Symptom Improvement

Confidence Interval Intuition

☆ What is the average amount (as measured by PHQ-9) a person with depression's symptoms change when they start taking this SSRI?

- Knowing that:

$$P\left(\bar{X} - 2\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 2\frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

- We can use the quantiles of the normal distribution at a pre-specified probability (here, 0.95) to get bounds for an interval around μ

- Creating an interval using this procedure has a 95% probability of capturing the population average amount that a person with depression's symptoms change when they start taking this SSRI

- If we repeatedly took random samples and built intervals this way, about 95% of those intervals would capture the true population mean

Summary

Topic 6: Sampling Distributions + CLT

- Statistics based on random samples are random variables with their own probability distributions, because they vary sample to sample
- Under certain conditions (i.i.d., large enough sample size) we can say that the distribution of certain statistics is approximately normal
 - This is called the central limit theorem, CLT

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \qquad \hat{P} \dot{\sim} N\left(p, \frac{p(1-p)}{n}\right)$$

- Using the CLT, we can work backwards from samples to try to get information (estimate + uncertainty, evidence for/against hypotheses) about population parameters
- Next topic, we'll talk about how to actually go about doing this